

OCR Further Mathematics
Discrete Mathematics
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



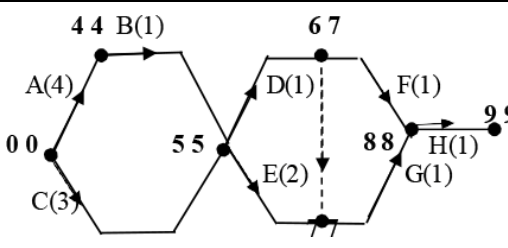
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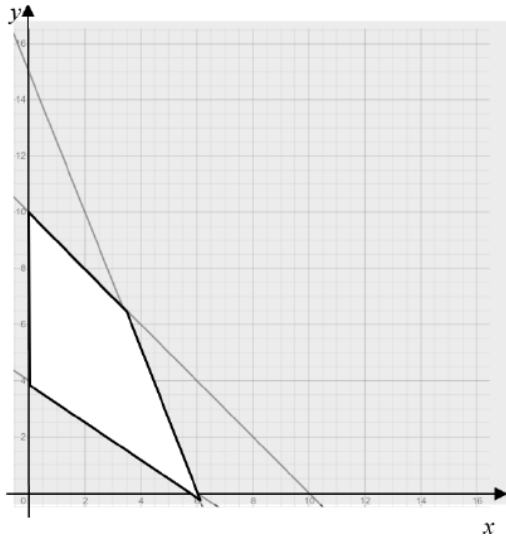
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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.



Question			Answer/Indicative content	Marks	Guidance
1	a	i	 Minimum time = 9 hours A, B, E, G, H have no float	M1 (A03.3) A1 (A01.1) M1 (A03.4) M1 (A03.4) A1 (A01.1) A1 (A01.1) [6]	Activity network with A, B and C correct D, E, F, G, H and dummy correct (accept directions missing) Forward pass attempted, or implied from min duration correct Backward pass attempted, or implied from critical activities correct 9 A, B, E, G, H (in any order) and no others
		ii	Assuming that there are enough workers for each activity Resourcing may restrict <u>how many</u> activities can happen together	B1 (A03.5b) [1]	A reason why it may not always be possible to do all the activities that are needed at the same time NOT an assumption about the durations or immediate predecessors or that would delay the start time of an activity (e.g. weather or delays in arrival of materials)
	b		Earliest time that E can start is 5 hours from start If there are not enough workers then A, B, C may need to be done one after another, taking 8 hours. And E could also be delayed until after D and F, giving a latest start time for E of 10 hours	B1 (A01.1) M1 (3.5a) A1 (A02.2b) [3]	5 (all the activities that <i>must</i> be done before E have min completion time 5) Recognising that tasks may be done sequentially (or implied from answer 8, 9 or 10) 10 (all the activities that <i>can</i> be done before E have total duration 10 – starting E after 10 would be an unnecessary delay)
	c		Extend the duration of D to 3 hours	B1 (A03.5c) [1]	Or add an activity immediately after D of duration 2 hours
			Total	11	



Question			Answer/Indicative content	Marks	Guidance															
2	a	i		M1 (AO 1.1) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [4]	Ignore any extra lines (e.g. profit lines) or working for parts (b), (c) Line $2x + 3y = 12$ through $(6, 0)$ and $(0, 4)$ Line $x + y = 10$ through at least two of $(10, 0)$, $(2, 8)$, $(4, 6)$, $(6, 4)$, $(8, 2)$ and $(0, 10)$ Line $5x + 2y = 30$ through at least two of $(6, 0)$, $(4, 5)$, $(2, 10)$ and $(0, 15)$ Feasible region identified and correct															
		ii	<table border="1"><thead><tr><th>x</th><th>y</th><th>$P = 4x - y$</th></tr></thead><tbody><tr><td>0</td><td>4</td><td>-4</td></tr><tr><td>0</td><td>10</td><td>-10</td></tr><tr><td>3.33</td><td>6.67</td><td>6.67</td></tr><tr><td>6</td><td>0</td><td>24</td></tr></tbody></table> Maximum $P = 24$	x	y	$P = 4x - y$	0	4	-4	0	10	-10	3.33	6.67	6.67	6	0	24	M1 (AO 3.1a) A1 (AO 1.1) [2]	'Determine' so method must be seen, not implied Checking at least two of their vertices or sliding a profit line (a line of gradient 4 anywhere on graph or indicating the vertex $(6, 0)$) 24
x	y	$P = 4x - y$																		
0	4	-4																		
0	10	-10																		
3.33	6.67	6.67																		
6	0	24																		
	b		FR has boundaries $x = 0$, $x + y = k$, $2x + 3y = 12$ $x + y = k$ and $2x + 3y = 12$ or $4x - y = 3$ Profit line $4x - y = 3$ cuts $2x + 3y = 12$ at $(1.5, 3)$ $k = 4.5$ Alternative solution $4x - (k - x) = 3 \Rightarrow 3x - k = 3$ and $2x + 3(k - x) = 12 \Rightarrow 3k - x = 12$ $k = 4.5$	M1 (AO3.4) M1 (AO3.1a) A1 (2.2a) Alternative solution M1 M1 A1 [3]	Not graphical Vertex where $2x + 3y = 12$ and $x + y = k$ or profit on line $x + y = k$ Calculate where profit = 3 on boundary $2x + 3y = 12$ or $(1.5, 3)$ 4.5 Alternative solution Use $x + y = k$ to substitute for y (or x) in $4x - y = 3$ Form a second simultaneous equation in the same unknowns 4.5															



Question			Answer/Indicative content	Marks	Guidance
	c		Profit line $4x - y = 3$ cuts $5x + 2y = 30$ at $\frac{3}{13}, \frac{1}{13}$ $k = \frac{141}{13}$	M1 (A03.1a) A1 (2.2a) [2]	Not vgraphical Or $2\frac{1}{13}, 8\frac{1}{13}$ or (2.7 to 2.8, 8.0 to 8.2) Or $10\frac{11}{13}$ or 10.8 to 10.9
			Total	11	



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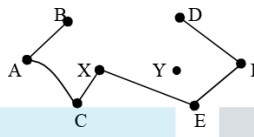
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Question			Answer/Indicative content	Marks	Guidance																																														
3	a	i	<table><tr><td colspan="2">Points won by Mia</td><td colspan="3">Mia</td></tr><tr><td></td><td></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td rowspan="3">Li</td><td>X</td><td>4</td><td>15</td><td>9</td></tr><tr><td>Y</td><td>11</td><td>6</td><td>5</td></tr><tr><td>Z</td><td>10</td><td>5</td><td>1</td></tr></table>	Points won by Mia		Mia					X	Y	Z	Li	X	4	15	9	Y	11	6	5	Z	10	5	1	B1 (AO 1.1) [1]	15, 9 and 6 all correct																							
Points won by Mia		Mia																																																	
		X	Y	Z																																															
Li	X	4	15	9																																															
	Y	11	6	5																																															
	Z	10	5	1																																															
		ii	<table><tr><td colspan="2">Points won by Li</td><td colspan="3">Mia</td></tr><tr><td></td><td></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td rowspan="3">Li</td><td>X</td><td>0.5</td><td>-10.5</td><td>-4.5</td></tr><tr><td>Y</td><td>-6.5</td><td>-1.5</td><td>-0.5</td></tr><tr><td>Z</td><td>-5.5</td><td>-0.5</td><td>3.5</td></tr></table> Alternative solution <table><tr><td colspan="2">Points won by Li</td><td colspan="3">Mia</td></tr><tr><td></td><td></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td rowspan="3">Li</td><td>X</td><td>1</td><td>-21</td><td>-9</td></tr><tr><td>Y</td><td>-13</td><td>-3</td><td>-1</td></tr><tr><td>Z</td><td>-11</td><td>-1</td><td>7</td></tr></table>	Points won by Li		Mia					X	Y	Z	Li	X	0.5	-10.5	-4.5	Y	-6.5	-1.5	-0.5	Z	-5.5	-0.5	3.5	Points won by Li		Mia					X	Y	Z	Li	X	1	-21	-9	Y	-13	-3	-1	Z	-11	-1	7	B1 (AO 1.1) Alternative solution B1 [1]	Subtract 4.5 from each value in the table showing points won by Li All correct Or any positive multiple of this table Alternative solution Points won by Li minus points won by Mia All correct (not follow through from (a))
Points won by Li		Mia																																																	
		X	Y	Z																																															
Li	X	0.5	-10.5	-4.5																																															
	Y	-6.5	-1.5	-0.5																																															
	Z	-5.5	-0.5	3.5																																															
Points won by Li		Mia																																																	
		X	Y	Z																																															
Li	X	1	-21	-9																																															
	Y	-13	-3	-1																																															
	Z	-11	-1	7																																															
	b		Row Z dominates row Y Column Y dominates column Z <table><tr><td colspan="2">Points won by Li</td><td colspan="2">Mia</td></tr><tr><td></td><td></td><td>X</td><td>Y</td></tr><tr><td rowspan="2">Li</td><td>X</td><td>0.5</td><td>-10.5</td></tr><tr><td>Z</td><td>-5.5</td><td>-0.5</td></tr></table>	Points won by Li		Mia				X	Y	Li	X	0.5	-10.5	Z	-5.5	-0.5	B1 (AO 1.1) B1 (AO 1.1) [2]	Row Y removed, seen in resulting 2 × 2 table Column Z removed, seen in resulting 2 × 2 table Follow through their table from part (a)(ii) or using original table <table><tr><td>e.g.</td><td>X</td><td>Y</td></tr><tr><td>X</td><td>1</td><td>-21</td></tr><tr><td>Z</td><td>-11</td><td>-1</td></tr></table> <table><tr><td>e.g.</td><td>X</td><td>Y</td></tr><tr><td>X</td><td>5</td><td>-6</td></tr><tr><td>Z</td><td>-1</td><td>4</td></tr></table>	e.g.	X	Y	X	1	-21	Z	-11	-1	e.g.	X	Y	X	5	-6	Z	-1	4													
Points won by Li		Mia																																																	
		X	Y																																																
Li	X	0.5	-10.5																																																
	Z	-5.5	-0.5																																																
e.g.	X	Y																																																	
X	1	-21																																																	
Z	-11	-1																																																	
e.g.	X	Y																																																	
X	5	-6																																																	
Z	-1	4																																																	
	c		Row minima are -10.5 and -5.5 Row maximin = -5.5 Li's play-safe is strategy Z Mia should play strategy X	M1 (AO 1.1) A1 ft (AO 1.1) B1 ft (AO 1.1) [3]	Follow through their table from part (b) or original tables Working may be seen on table from part (b) Seen or implied from row minima, or identifying play-safe for Li Or indicated in table, or implied from their row maximin value stated Correct choice for their table from part (b)																																														
			Total	7																																															



Question			Answer/Indicative content	Marks	Guidance																																				
4	a		ABXE	B1 (AO 1.1) [1]																																					
	b		<table border="1"> <tr><td>A</td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>0.6</td><td>B</td><td></td><td></td><td></td><td></td></tr> <tr><td>1.1</td><td>1.7</td><td>C</td><td></td><td></td><td></td></tr> <tr><td>2.7</td><td>2.1</td><td>2.5</td><td>D</td><td></td><td></td></tr> <tr><td>2.8</td><td>2.2</td><td>1.8</td><td>1.2</td><td>E</td><td></td></tr> <tr><td>3.3</td><td>2.7</td><td>2.5</td><td>0.6</td><td>0.7</td><td>F</td></tr> </table>	A						0.6	B					1.1	1.7	C				2.7	2.1	2.5	D			2.8	2.2	1.8	1.2	E		3.3	2.7	2.5	0.6	0.7	F	M1 (AO 1.1) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [4]	AB = 0.6, AC = 1.1 AD = 2.7, AE = 2.8 DF = 0.6, EF = 0.7 BF = 2.7, CF = 2.5 AF = 3.3 or ft from other values
A																																									
0.6	B																																								
1.1	1.7	C																																							
2.7	2.1	2.5	D																																						
2.8	2.2	1.8	1.2	E																																					
3.3	2.7	2.5	0.6	0.7	F																																				
	c		AB = 0.6 AC = 1.1 CE = 1.8 EF = 0.7 DF = <u>0.6</u> 4.8 	M1 (AO 3.1b) A1 (AO 3.2a) B1 ft (AO 1.1) [3]	A graph that connects {A, B, C, D, E, F} with or without X and / or Y Correct tree drawn or arcs listed, including CX and XE 4.8 (km) or total for their tree																																				
	d		Adapting the answer to part (c) B - A - C - X - Y - E - F - D	M1 (AO 3.1b) A1 (AO 1.1) [2]	Any walk or cycle that starts at B and uses every vertex at least once, including X and Y cao																																				
			Total	10																																					



Question			Answer/Indicative content	Marks	Guidance																									
5	a	i	Next-fit method <table><tr><td>Bin 1</td><td>12</td><td></td><td></td><td></td></tr><tr><td>Bin 2</td><td>23</td><td></td><td></td><td></td></tr><tr><td>Bin 3</td><td>15</td><td></td><td></td><td></td></tr><tr><td>Bin 4</td><td>18</td><td>8</td><td></td><td></td></tr><tr><td>Bin 5</td><td>7</td><td>5</td><td></td><td></td></tr></table>	Bin 1	12				Bin 2	23				Bin 3	15				Bin 4	18	8			Bin 5	7	5			M1 (AO 1.1) A1 (AO 1.1) [2]	Bins 1 and 2 correct All correct
Bin 1	12																													
Bin 2	23																													
Bin 3	15																													
Bin 4	18	8																												
Bin 5	7	5																												
		ii	First-fit method <table><tr><td>Bin 1</td><td>12</td><td>15</td><td></td><td></td></tr><tr><td>Bin 2</td><td>23</td><td>7</td><td></td><td></td></tr><tr><td>Bin 3</td><td>18</td><td>8</td><td></td><td></td></tr><tr><td>Bin 4</td><td>5</td><td></td><td></td><td></td></tr><tr><td>Bin 5</td><td></td><td></td><td></td><td></td></tr></table>	Bin 1	12	15			Bin 2	23	7			Bin 3	18	8			Bin 4	5				Bin 5					M1 (AO 1.1) A1 (AO 1.1) [2]	Bins 1 and 2 correct All correct
Bin 1	12	15																												
Bin 2	23	7																												
Bin 3	18	8																												
Bin 4	5																													
Bin 5																														
		iii	23 18 15 12 8 7 5 First-fit decreasing method <table><tr><td>Bin 1</td><td>23</td><td>7</td><td></td><td></td></tr><tr><td>Bin 2</td><td>18</td><td>12</td><td></td><td></td></tr><tr><td>Bin 3</td><td>15</td><td>8</td><td>5</td><td></td></tr><tr><td>Bin 4</td><td></td><td></td><td></td><td></td></tr><tr><td>Bin 5</td><td></td><td></td><td></td><td></td></tr></table>	Bin 1	23	7			Bin 2	18	12			Bin 3	15	8	5		Bin 4					Bin 5					M1 (AO 1.1) A1 (AO 1.1) [2]	Ordered list may be seen Bins 1 and 2 correct All correct
Bin 1	23	7																												
Bin 2	18	12																												
Bin 3	15	8	5																											
Bin 4																														
Bin 5																														
	b		With ‘online’ lists the items are presented one at a time and the whole list is not known until the end. With next-fit and first-fit the items are placed in the order they appear in the list, so these methods can be used ‘offline’ or ‘online’. However, for first-fit decreasing the whole list needs to be known before it can be sorted, so first-fit decreasing can only be used for an ‘offline’ list.	B1 (AO 1.2) B1 (AO 2.3) [2]	Evidence of understanding what ‘online’ means Evidence of realising that ffd cannot be used with an online list (or implied from an appropriate statement about next-fit and first-fit)																									



Question			Answer/Indicative content	Marks	Guidance
	c		$88 \div 4 = 22$, so M is at least 22 But it is not possible to fill 4 bins of capacity 22 Since $22 - 18 = 4$ which is less than 5 So the 23 would have to be split as 4 and 19 And then there is no 3 to go with the 19 $M = 23$ is possible e.g. $23 - x$ and x, $18 + 5$, $15 + 8$, $12 + 7$ Hence, least M is 23	M1 (AO 1.1) A1 (AO 2.4) B1 (AO 2.2a) [3]	Identifying that M must be at least 22 Showing that $M = 22$ is not possible Fully correct explanation Showing that $M = 23$ is possible
			Total	11	



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Question		Answer/Indicative content	Marks	Guidance
6	a	5 partitions into a set of size 1 and a set of size 4 $\{X \mid X, X, X, X\}$ 10 partitions into a set of size 2 and a set of size 3 $\{X, X \mid X, X, X\}$ because there are 5C_2 choices for the set of size 2 Alternative solution $\{A\}, \{B, C, D, E\} \quad \{B\}, \{A, C, D, E\}$ $\{C\}, \{A, B, D, E\} \quad \{D\}, \{A, B, C, E\}$ $\{E\}, \{A, B, C, D\}$ $\{A, B\}, \{C, D, E\} \quad \{A, C\}, \{B, D, E\}$ $\{A, D\}, \{B, C, E\} \quad \{A, E\}, \{B, C, D\}$ $\{B, C\}, \{A, D, E\} \quad \{B, D\}, \{A, C, E\}$ $\{B, E\}, \{A, C, D\} \quad \{C, D\}, \{A, B, E\}$ $\{C, E\}, \{A, B, D\} \quad \{D, E\}, \{A, B, C\}$	B1 (AO 1.1) B1 (AO 2.5) Alternative solution B1 B1 [2]	5 where smaller set has size 1 or ${}^5C_1 = 5$ 10 where smaller set has size 2, with an explanation of why it is 10 (note the total of 15 is given in the question) e.g. ${}^5C_2 = 10$ or $(5 \times 4) \div 2 = 10$ or $4 + 3 + 2 + 1 = 10$ Alternative solution List (or any equivalent) that has exactly 5 distinct cases where smaller set has size 1 May just list one set, e.g. $\{A\}, \{B\}, \{C\}, \{D\}, \{E\}$ List (or any equivalent) that has 10 distinct cases sets where smaller set has size 2 May just list one set, e.g. $\{A, B\}, \{A, C\}, \{A, D\}, \{A, E\}, \{B, C\}, \{B, D\}, \{B, E\}, \{C, D\}, \{C, E\}, \{D, E\}$
	b	Partitions into sets of sizes 1, 1 and 3 $5 \times 4 \div 2 = 10$ partitions of this type Partitions into sets of sizes 1, 2 and 2 $5 \times ({}^4C_2 \div 2) = 5 \times 3 = 15$ partitions of this type	M1 (AO 1.1) A1 (AO 2.1) M1 (AO 1.1) A1 (AO 2.1) [4]	Considering cases where set sizes are 1, 1, 3 Explanation of why there are 10 of these e.g. ${}^5C_3 = 10$ or $5 \times 4 \div 2 = 10$ or a list of the cases Considering cases where set sizes are 1, 2, 2 Explaining why there are 15 of these e.g. a relevant calculation or list of cases
		10 partitions into sets of sizes 1, 1, 1, 2 1 partition into sets of sizes 1, 1, 1, 1, 1 $15 + 25 + 10 + 1 = 51$	M1 (AO 2.1) A1 (AO 1.1) [2]	Trying to deal with the cases when there are more than 3 subsets May be implied from answer 51 51
	d	Number line is split into 6 pieces But there are 8 numbers Hence result by the pigeonhole principle	B1 (AO 2.1) B1 (AO 2.2a) [2]	6 pieces Using pigeonhole, or explaining why there must be at least one piece with two or more numbers
		Total	10	

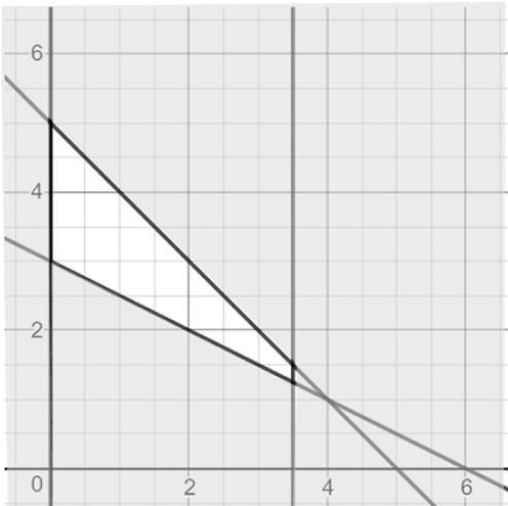


Question			Answer/Indicative content	Marks	Guidance
7	a		$6 - x - y$	B1 [1]	This expression, in any form
	b	i	Total value = $5x + 3y + 2(6 - x - y)$ $= 3x + y + 12$ 12 is constant so total value is max when $3x + y$ is max	M1 M1 A1 [3]	$5x + 3y$ + $2(6 - x - y)$ or 2(their expression from part (a)) $3x + y + 12$ seen or explaining why constant can be ignored
		ii	$y \geq 6 - x - y$ $\Rightarrow x + 2y \geq 6$	M1 A1 [2]	$y \geq$ their expression from part (a) Leading to given expression
		iii	$x \leq 3.5$ and $x + 2y \geq 6$ $\Rightarrow y \geq 1.25$ (or $y \geq 0$)	M1 A1 [2]	x is at most 3.5 (or x is at most 5) and $x + 2y$ is at least 6 Leading to implication that y cannot be negative (as given) Argued algebraically or in words

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Question		Answer/Indicative content	Marks	Guidance															
c				May have other lines shown, e.g. profit lines Assume x is on the horizontal axis and y on the vertical axis, unless labelled to the contrary															
			M1	Vertical boundary $x = 3.5$															
			M1	Line $x + 2y = 6$ through $(0, 3)$ and $(4, 1)$															
			M1	Line $x + y = 5$, through $(0, 5)$ and $(4, 1)$															
			A1	Shading to show this feasible region															
		<table border="1" data-bbox="300 857 807 990"> <tr> <td>x</td><td>0</td><td>3.5</td><td>3.5</td><td>0</td></tr> <tr> <td>y</td><td>3</td><td>1.25</td><td>1.5</td><td>5</td></tr> <tr> <td>$3x + y$</td><td>3</td><td>11.75</td><td>12</td><td>5</td></tr> </table> Tamsin should spend 3.5 hours walking the coast path 1.5 hours visiting the bird sanctuary 1 hour visiting the garden centre	x	0	3.5	3.5	0	y	3	1.25	1.5	5	$3x + y$	3	11.75	12	5	M1 M1 A1 [7]	Checking vertices or using a sliding profit line May be implied from their solution provided consistent with their feasible region Interpretation of their x and y in context, provided $x \leq 3.5$ and both are ≥ 0 $(6 - \text{their } x - \text{their } y)$ hours (≥ 0) visiting the garden centre
x	0	3.5	3.5	0															
y	3	1.25	1.5	5															
$3x + y$	3	11.75	12	5															
		Total	15																

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Question			Answer/Indicative content	Marks	Guidance
8	a		5	B1 [1]	cao
	b		Row minima: $A = \min\{2k, -2, 1 - k\}$ $B = -5$ $C = -4$ $D = -5$ $2k = -4$ when $k = -2$ $1 - k = -4$ when $k = 5$ So A is a play-safe strategy when $-2 \leq k \leq 5$	M1 A1 B1 B1 M1 A1 [6]	$2k, -2, 1 - k$ (may also include 4) -4 seen or used appropriately Critical value $k = -2$ (or as an inequality) Critical value $k = 5$ (or as an inequality) Appropriate sketch graph showing $y = 2k$, $y = 1 - k$ and at least one of $y = -2$, $y = -4$ Or implied from answer Accept $<$ instead of (either or both) $\leq$
	c		E: $\max\{2k, 4\}$, F: 4 G: $\max\{4, 1 - k\}$, H: 6 $\max\{2k, 4\} \geq 4$ and $\max\{4, 1 - k\} \geq 4$ and $F = 4$ Hence col minimax = 4	M1 A1 [2]	Any three of these correct (not implied), 'max' may be implied 4 from correct working seen
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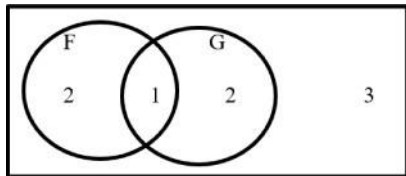


Question		Answer/Indicative content	Marks	Guidance
9	a	Minimum project completion time = 60 days Critical activities: A, B, C, G, H, I, K	M1	H, I, J, K added appropriately, apart from possibly not using a dummy (or using too many dummies) and possibly not showing arcs as directed. Durations need not be shown.
			A1	Network correct, including the use of exactly one dummy (before or after I or J) and all arcs directed
			M1	Follow through their network, even if only A to G
			A1	Forward pass, or implied from minimum time correct 60
			M1	Backward pass, or implied from critical activities correct
			A1 [6]	A, B, C, G, H, I, K (and no others)
	b	The start of an activity may be delayed (e.g. a delay in the delivery would hold up the start of G or an activity may take longer than Bob has planned)	B1	A reason why the duration of an activity may be extended
		There may not be enough workers available who can do the activities at the required times	B1 [2]	A resourcing issue related to the worker availability
	c	F has the longest float = $25 - 1 - 5 = 19$ days The clearing of the work area can start at any time from the end of A (day 1) to day 20 without delaying the project	M1 A1 [2]	19 or ft their forward and backward passes Interpretation in context, must refer to 'clearing work area' and start or finish times (or needing 5 days from the window of 24 days) or reference to opportunity window
Total			10	



Question			Answer/Indicative content	Marks	Guidance																																																																																
10	a	i	e.g. A – B – A – D – H e.g. A – B – C – D – E – C – D – H e.g. A – D – E – D – H	B1 [1]	A walk from A to H that repeats an arc (or arcs)																																																																																
		ii	e.g. A – D – C – E – D – H e.g. A – B – C – D – A – G – H	B1 [1]	A trail from A to H (does not repeat any arcs) that does repeat a node (or nodes)																																																																																
		iii	e.g. A – D – H e.g. A – B – C – D – E – G – H	B1 [1]	A path from A to H (does not repeat any nodes)																																																																																
	b		<table border="1"><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th></tr></thead><tbody><tr><th>A</th><td>-</td><td>3</td><td>-</td><td>8</td><td>-</td><td>-</td><td>x</td><td>-</td></tr><tr><th>B</th><td>3</td><td>-</td><td>5</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><th>C</th><td>-</td><td>5</td><td>-</td><td>4</td><td>3</td><td>-</td><td>-</td><td>-</td></tr><tr><th>D</th><td>8</td><td>-</td><td>4</td><td>-</td><td>5</td><td>-</td><td>-</td><td>9</td></tr><tr><th>E</th><td>-</td><td>-</td><td>3</td><td>5</td><td>-</td><td>7</td><td>6</td><td>-</td></tr><tr><th>F</th><td>-</td><td>-</td><td>-</td><td>-</td><td>7</td><td>-</td><td>-</td><td>-</td></tr><tr><th>G</th><td>x</td><td>-</td><td>-</td><td>-</td><td>6</td><td>-</td><td>-</td><td>2</td></tr><tr><th>H</th><td>-</td><td>-</td><td>-</td><td>9</td><td>-</td><td>-</td><td>2</td><td>-</td></tr></tbody></table> AB = 3 BC = 5 CE = 3 CD = 4 EG = 6 GH = 2 EF = 7		A	B	C	D	E	F	G	H	A	-	3	-	8	-	-	x	-	B	3	-	5	-	-	-	-	-	C	-	5	-	4	3	-	-	-	D	8	-	4	-	5	-	-	9	E	-	-	3	5	-	7	6	-	F	-	-	-	-	7	-	-	-	G	x	-	-	-	6	-	-	2	H	-	-	-	9	-	-	2	-	B1 M1 A1 For reference Choosing the 7 circled entries, shown on table Must be on the table and all correct (not any transposes) A spanning tree for the 8 nodes need not match the arcs in the network, need not be MST Correct tree drawn or correct arcs listed (not just on table)
	A	B	C	D	E	F	G	H																																																																													
A	-	3	-	8	-	-	x	-																																																																													
B	3	-	5	-	-	-	-	-																																																																													
C	-	5	-	4	3	-	-	-																																																																													
D	8	-	4	-	5	-	-	9																																																																													
E	-	-	3	5	-	7	6	-																																																																													
F	-	-	-	-	7	-	-	-																																																																													
G	x	-	-	-	6	-	-	2																																																																													
H	-	-	-	9	-	-	2	-																																																																													
	c	i	{x: 0 ≤ x ≤ 6}	B1 [1]	x ≤ 6, x ≤ (their largest weight used to A or G) Allow strict inequality																																																																																
		ii	EG	B1 [1]	Follow through their spanning tree from (b)																																																																																
			Total	8																																																																																	



Question			Answer/Indicative content	Marks	Guidance										
11	a		How can I fit the packages in the boxes? She needs to find a solution, although it may not be a good solution She does not need to enumerate the number of solutions	B1 [1]	Correct question chosen with a valid reason										
	b		A solution may not exist because the packages may be the wrong shapes. May have a package that is too big ($> 0.25 \text{ m}^3$) Could have packages that cannot be paired together to fit in six boxes (e.g. $6 \times 0.15 \text{ m}^3$ and $4 \times 0.025 \text{ m}^3$)	B1 [1]	Shape matters as well as volume Or explaining why total volume = 1 m^3 does not mean that individual volumes are small enough										
	c		Would need all four boxes full, with no space left A(0.20), C(0.15), D(0.25) must be in 3 separate boxes. I(0.12) and J(0.12) cannot be in the same box as A, C or D so I and J are together in another box The box with I and J has 0.01 spare, but the smallest package has volume 0.02. So impossible to fill 4 boxes	B1 B1 [2]	This first statement may be implied I and J must be together (e.g showing a solution in which A, C and D are in separate boxes <u>and</u> I, J are together in a box) Explaining why this means that 4 boxes are not enough (not just showing that first fit decreasing fails)										
	d		<table><tr><td>Box 1</td><td>A B</td></tr><tr><td>Box 2</td><td>C E F G</td></tr><tr><td>Box 3</td><td>D</td></tr><tr><td>Box 4</td><td>H I</td></tr><tr><td>Box 5</td><td>J</td></tr></table>	Box 1	A B	Box 2	C E F G	Box 3	D	Box 4	H I	Box 5	J	M1 A1 [2]	A, B, C, D placed correctly (if letters are not used, 0.20, 0.05, 0.15, 0.25 placed correctly = M1 A0) All correct, using letters
Box 1	A B														
Box 2	C E F G														
Box 3	D														
Box 4	H I														
Box 5	J														
	e		$8 - (3 + 3) + 1 = 3$	M1 A1 [2]	Or $3 + 3 - 1 = 5$ shown on a Venn diagram 3, from correct working seen 										
			Total	8											



Question		Answer/Indicative content	Marks	Guidance																																																																																																								
12	a	<table><tr><td>A</td><td>0</td><td></td><td></td><td></td><td>8</td><td></td><td></td><td></td><td>-1</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>5</td><td></td><td></td><td></td><td>4</td><td></td><td></td><td></td><td>3</td><td></td><td></td></tr><tr><td>C</td><td></td><td></td><td>1</td><td></td><td></td><td></td><td>1</td><td></td><td></td><td></td><td>1</td><td></td></tr><tr><td>D</td><td></td><td></td><td></td><td>8</td><td></td><td></td><td></td><td>7</td><td></td><td></td><td></td><td>6</td></tr></table> <table><tr><td>A</td><td>7</td><td></td><td></td><td></td><td>-5</td><td></td><td></td><td></td><td>6</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>2</td><td></td><td></td><td></td><td>1</td><td></td><td></td><td></td><td>0</td><td></td><td></td></tr><tr><td>C</td><td></td><td></td><td>0</td><td></td><td></td><td></td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>D</td><td></td><td></td><td></td><td>2</td><td></td><td></td><td></td><td>1</td><td></td><td></td><td></td><td></td></tr></table> Output = 6	A	0				8				-1				B		5				4				3			C			1				1				1		D				8				7				6	A	7				-5				6				B		2				1				0			C			0				0						D				2				1					B1 B1 M1 A1 B1 [5]	Allow compressed rows or tables used differently Values of B are 5, 4, 3, 2, 1, 0 Values of C are 1, 1, 1, 0, 0 Ignore if there is an extra 0 at the end Their first two D values = $B + 3C$ All D values correct: 8, 7, 6, 2, 1 Ignore if there is an extra 0 at the end 6
A	0				8				-1																																																																																																			
B		5				4				3																																																																																																		
C			1				1				1																																																																																																	
D				8				7				6																																																																																																
A	7				-5				6																																																																																																			
B		2				1				0																																																																																																		
C			0				0																																																																																																					
D				2				1																																																																																																				
	b	Value of B is reduced by 1 each time that STEP 6 is used until it reaches 0, when STEP 7 and STEP 8 mean that the algorithm stops	B1 B1 [2]	B reduces by 1 in each pass (NOT just B is a counter) B reaches 0 or referring to STEP 7 and / or STEP 8 or STOP instruction																																																																																																								
	c	For any given input the output is always the same.	B1 [1]	The output is a function of the input only																																																																																																								
	d	B reduces to 0 after n passes through STEP 6, so run time is kn which is a linear function of n Hence efficiency is $O(n)$ (as given in question)	B1 [1]	Identifying n passes (allow $n \pm 1$) Or ' B reduces by 1 in each pass' so (run time) is linear, or equivalent																																																																																																								
	e	For large enough n , the run time of X increases more rapidly than the runtime of Y . Alternative method The number of digits in B is $1 + \text{INT}(\log_{10} B)$. For any positive values of k and m there will be a value N for which the run-time of $X >$ the run time of Y for all $n > N$.	B1 B1 [1]	Referring to 'for large values of n ' (or B) and not just saying that the number of digits is smaller than n . Not just specific examples. Any reasonable attempt at number of digits in B that involves logarithms A level candidates may give $O(\log n) < O(n)$ Need not explain this																																																																																																								
		Total	10																																																																																																									

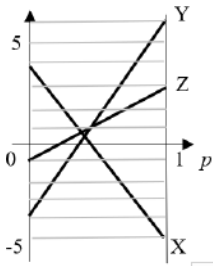


Question			Answer/Indicative content	Marks	Guidance																																								
13	a		Total pay-off is 16, subtract 8 from each pay-off. <table><tr><td></td><td></td><td colspan="3">Emma</td></tr><tr><td></td><td></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td rowspan="3">Drew</td><td>P</td><td>-5</td><td>6</td><td>3</td></tr><tr><td>Q</td><td>4</td><td>-4</td><td>-1</td></tr><tr><td>R</td><td>3</td><td>-4</td><td>-2</td></tr></table>			Emma					X	Y	Z	Drew	P	-5	6	3	Q	4	-4	-1	R	3	-4	-2	M1(A01.1) A1(A01.1)	Converting to zero-sum, or implied from total = 16 (for each cell) or from working Pay-off matrix for Drew all correct (or a positive multiple of this)	Each player puts 8 into 'pot' May use differences <table><tr><td></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>P</td><td>-10</td><td>12</td><td>6</td></tr><tr><td>Q</td><td>8</td><td>-8</td><td>-2</td></tr><tr><td>R</td><td>6</td><td>-8</td><td>-4</td></tr></table>		X	Y	Z	P	-10	12	6	Q	8	-8	-2	R	6	-8	-4
		Emma																																											
		X	Y	Z																																									
Drew	P	-5	6	3																																									
	Q	4	-4	-1																																									
	R	3	-4	-2																																									
	X	Y	Z																																										
P	-10	12	6																																										
Q	8	-8	-2																																										
R	6	-8	-4																																										

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Question		Answer/Indicative content	Marks	Guidance																					
b		Strategy Q (weakly) dominates strategy R so matrix can be reduced <table><tr><td colspan="2"></td><td colspan="3">Emma</td></tr><tr><td colspan="2"></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>P</td><td></td><td>-5</td><td>6</td><td>3</td></tr><tr><td>Q</td><td></td><td>4</td><td>-4</td><td>-1</td></tr></table>			Emma					X	Y	Z	P		-5	6	3	Q		4	-4	-1	M1(A01.1)	ANSWER SPACE IS ON 2 PAGES Identifying dominance Row (from their zero-sum table) removed by dominance	If an A-level candidate chooses to use simplex credit as below: M1: Add 5 to make all entries non-negative, set up initial tableau A1: Initial tableau correct
			Emma																						
			X	Y	Z																				
	P		-5	6	3																				
	Q		4	-4	-1																				
	Let Drew choose randomly between P and Q so that the probability of P is p and the probability of Q is $1 - p$.	A1(A02.1)	Describing (prob of) $P = p$ and (prob of) $Q = 1 - p$, or equivalent, in terms of <u>one</u> variable only Or $4 - 9p$	B1: At least two valid iterations M1 Achieving final tableau																					
	If Emma plays X, Drew expects $-5p + 4(1 - p)$ If Emma plays Y, Drew expects $6p - 4(1 - p)$ If Emma plays Z, Drew expects $3p - (1 - p)$	B1(A01.1)	Or $10p - 4$ Or $4p - 1$ All three correct (in any form) for the values in their 2×3 reduced table Or implied from graph if no expressions are given	with $p = \frac{8}{19}$, $q = \frac{11}{19}$, $r = 0$ A1 Correct interpretation in context.																					
				Check with TL if necessary																					
	$4 - 9p = 10p - 4$ $p = \frac{8}{19}$	M1(A01.1)	Solving simultaneous equations when $X = Y$																						
	P with probability $\frac{8}{19}$ and Q with probability $\frac{11}{19}$	A1(A02.1)	Both probabilities correct in context																						



Question			Answer/Indicative content	Marks	Guidance	
	c		From graph, if Drew plays optimally then Emma should not choose Z. Let Emma choose randomly between X and Y so that the probability of X is x and the probability of Y is $1 - x$. $5x - 6(1 - x) = -4x + 4(1 - x)$ or $= -\frac{4}{19}$ X with probability $\frac{10}{19}$ and Y with probability $\frac{9}{19}$	M1(A02.2a) A1(A02.1) B1(A01.1) M1(A01.1) A1(A02.1) [12]	Or equivalent Description including symbols used for probabilities (one unknown only) Pay-offs for Emma are $\times (-1)$ A valid method Either probability correct	
			Total	12		



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Question			Answer/Indicative content	Marks	Guidance	
14	a		g, r, s = time spent on gym work, running, swimming, respectively, in hours Maximise $3g + r + 1.5s$ subject to $g + r + s \leq 18$ $g \leq 8$ $r \geq 4$ $g + s \leq r$ $g \geq 0$ and $s \geq 0$	B1(A01.1) B1(A03.3) B1(A01.1) B1(A03.3) B1(A01.1)	Define variables Objective (maximise and positive coefficients in ratio 6:2:3) $g + r + s \leq 18$ $g + s \leq r$ $0 \leq g \leq 8, s \geq 0$ and $r \geq 4$	Or equivalent with other letters e.g. x, y, z Max $g + \frac{1}{3}r + \frac{1}{2}s$ NOT Max $g + 3r + 2s$ Ignore extra constraints provided they are not inconsistent
	b	i	Set $g = 8$: Maximise $r + 1.5s$ (+24) subject to $r + s \leq 10, s + 8 \leq r$ and $s \geq 0$ e.g. Optimal at $r = 9, s = 1$ Spend 9 hours running and 1 hour swimming.	B1(A03.1b) M1(A01.1) A1(A03.4) M1(A01.1) A1(A03.4)	ANSWER SPACE IS ON 2 PAGES Reformulate as a 2-variable problem with an objective function and at least two non-trivial constraints A finite FR identified on graph with variables their r and s Correct FR identified A vertex where 2 of their lines (with variables their r and s) cross Interpret their solution (with non-negative times) in context	A linear expression or linear equation and at least two non-trivial constraints in their r and s M1 may be implied from correct solution in context
		ii	e.g. would need to rest / eat/sleep e.g. need transition time between activities	B1(A03.5b)	A reason <u>why</u> their solution is not practical	e.g. should not train for 18 hours just before a race



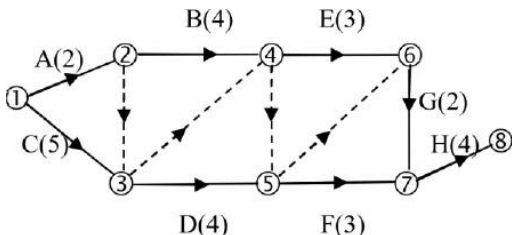
Question			Answer/Indicative content	Marks	Guidance	
		iii	e.g. build in recovery time e.g. for each hour allocated to each type of training only 50 minutes is spent on the training.	B1(A03.5c) [12]	Describing <u>how</u> model can be adapted to make their solution more realistic (do not need specific times)	e.g. restrict the preparation time to something that is less than 18 hours
			Total	12		



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Question			Answer/Indicative content	Marks	Guidance																																																							
15	a		 <table border="1" data-bbox="298 645 804 844"><tr><td>Event</td><td>①</td><td>②</td><td>③</td><td>④</td><td>⑤</td><td>⑥</td><td>⑦</td><td>⑧</td></tr><tr><td>Early time</td><td>0</td><td>2</td><td>5</td><td>6</td><td>9</td><td>9</td><td>12</td><td>16</td></tr><tr><td>late time</td><td>0</td><td>3</td><td>5</td><td>7</td><td>9</td><td>10</td><td>12</td><td>16</td></tr></table> Longest path C – D – F – H <table border="1" data-bbox="298 1108 804 1308"><tr><td>Event</td><td>①</td><td>②</td><td>③</td><td>④</td><td>⑤</td><td>⑥</td><td>⑦</td><td>⑧</td></tr><tr><td>Temporary</td><td></td><td>2</td><td>5</td><td>6</td><td>6</td><td>5</td><td>5</td><td>8</td></tr><tr><td>Permanent</td><td>0</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>4</td><td>8</td></tr></table> Shortest path A – G – H	Event	①	②	③	④	⑤	⑥	⑦	⑧	Early time	0	2	5	6	9	9	12	16	late time	0	3	5	7	9	10	12	16	Event	①	②	③	④	⑤	⑥	⑦	⑧	Temporary		2	5	6	6	5	5	8	Permanent	0	2	2	2	2	2	4	8	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1ft(AO1.1) B1(AO3.2a) M1(AO1.1) A1ft(AO1.1) B1(AO3.2a)	ANSWER SPACE IS ON 2 PAGES Activity on arc network with 8 activities, A to H Correct, with directed arcs and no extra dummies. Durations need not be shown Forward pass attempted Forward and backward passes correct, follow through their activity on arc network with at least one burst and at least one merge Critical path correct Evidence of Dijkstra (at vertices) Follow through their network with at least one vertex updated Shortest route correct	M1 may be implied from 16 as length of longest path (www) or from correct longest path Or equivalent description M1 may be implied from 8 as length of shortest path (www) or from correct shortest path Or equivalent description
Event	①	②	③	④	⑤	⑥	⑦	⑧																																																				
Early time	0	2	5	6	9	9	12	16																																																				
late time	0	3	5	7	9	10	12	16																																																				
Event	①	②	③	④	⑤	⑥	⑦	⑧																																																				
Temporary		2	5	6	6	5	5	8																																																				
Permanent	0	2	2	2	2	2	4	8																																																				
	b		Minimum completion time	B1(AO3.2a)	Or equivalent	Referring to time for critical activities (or equivalent)																																																						
	c		1	B1(AO3.2a) [10]	Units may be implied																																																							
Total				10																																																								



Question			Answer/Indicative content	Marks	Guidance																																									
16	a		Shuttle (sort)	B1(A01.2)	Or insertion																																									
	b		<table><tr><td>Input list</td><td>L</td><td>N</td><td>Output list</td></tr><tr><td>5 7 6 4</td><td>6</td><td>9</td><td>6</td></tr><tr><td>7 6 4</td><td>6</td><td>5</td><td>6 9</td></tr><tr><td>6 4</td><td>5</td><td>7</td><td>5 6 9</td></tr><tr><td>6 4</td><td>6</td><td>7</td><td>5 6 9</td></tr><tr><td>6 4</td><td>9</td><td>7</td><td>5 6 9</td></tr><tr><td>4</td><td>5</td><td>6</td><td>5 6 7 9</td></tr><tr><td>4</td><td>6</td><td>6</td><td>5 6 7 9</td></tr><tr><td></td><td>5</td><td>4</td><td>5 6 6 7 9</td></tr><tr><td></td><td></td><td></td><td>4 5 6 6 7 9</td></tr></table>	Input list	L	N	Output list	5 7 6 4	6	9	6	7 6 4	6	5	6 9	6 4	5	7	5 6 9	6 4	6	7	5 6 9	6 4	9	7	5 6 9	4	5	6	5 6 7 9	4	6	6	5 6 7 9		5	4	5 6 6 7 9				4 5 6 6 7 9	B1(A01.1) B1(A01.1) B1(A01.1) B1(A01.1) B1(A01.1)	ANSWER SPACE IS ON 2 PAGES First row correct in all four columns L = 6, N = 9 and then N = 5 L = 5, 6, 9 (in this order) with N = 7 L = 5, 6 (in this order) with N = 6 {input list} U {N} U {output list} = {4, 5, 6, 6, 7, 9} in at least 4 different rows	Treat any blank cells with values on that column in a row above and a row below as rolling down from the row above
Input list	L	N	Output list																																											
5 7 6 4	6	9	6																																											
7 6 4	6	5	6 9																																											
6 4	5	7	5 6 9																																											
6 4	6	7	5 6 9																																											
6 4	9	7	5 6 9																																											
4	5	6	5 6 7 9																																											
4	6	6	5 6 7 9																																											
	5	4	5 6 6 7 9																																											
			4 5 6 6 7 9																																											
	c	i	8	B1ft(A01.1)	Correct or follow through above part	Number of different rows, excluding 'print' row																																								
		ii	15	B1(A02.2b)	1+2+3+4+5																																									
	d		Sorting is $O(n^2)$ or quadratic order $15 \times \left(\frac{300}{60}\right)^2$ = 375 seconds	M1(A01.1) M1(A01.1) A1(A02.2b) [11]	Seen or implied from working Size of problem is $\times \frac{300}{60} = \times 5$ Or $k \times 60^2 = 15$ so $k = \frac{1}{240}$ cao, with units	Allow M1 for $\frac{1}{2}n(n-1)$ or any quadratic function, algebraic M0, M1 for treating problem as $O(n)$ or $O(n^3)$ Or 6 minutes 15 seconds																																								
			Total	11																																										

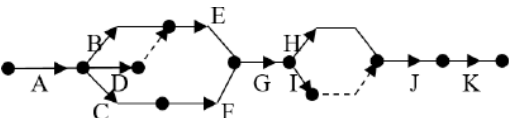


Question			Answer/Indicative content	Marks	Guidance																									
17	a		<table><tr><td>Vertex</td><td>J</td><td>K</td><td>L</td><td>M</td><td>N</td></tr><tr><td>Degree</td><td>1</td><td>3</td><td>2</td><td>2</td><td>2</td></tr></table> <table><tr><td>Vertex</td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td></tr><tr><td>Degree</td><td>3</td><td>2</td><td>2</td><td>2</td><td>1</td></tr></table>	Vertex	J	K	L	M	N	Degree	1	3	2	2	2	Vertex	P	Q	R	S	T	Degree	3	2	2	2	1	M1(A01.1) M1(A01.1) A1(A01.1)	Each graph has at least three vertices of degree 2 listed J and T each have degree 1 All correct	
Vertex	J	K	L	M	N																									
Degree	1	3	2	2	2																									
Vertex	P	Q	R	S	T																									
Degree	3	2	2	2	1																									
	b		e.g. In G1, the vertex with degree 3 is connected to the vertex of degree 1 (and two of degree 2) In G2, the vertex of degree 3 is connected to three vertices of degree 2 e.g. In G1 the vertex of degree 1 is connected to the vertex of degree 3 but in G2 it is connected to a vertex of degree 2.	B1(A01.1) B1(A02.1)	The region in G1 has four edges (regions have four edges) The region in G2 has three edges (regions have three edges) Or equivalent <u>written</u> explanation	Incomplete explanation that uses both graphs = B1, B0 e.g. No arc between J and N but P is joined to Q																								
	c	i	Still (at least one) odd degree	B1(A02.1)	Still have vertices of degree 3	P and/or K not even degrees																								
		ii	Add an arc joining P to K	B1(A01.1)	P to K, written	Arc PK																								
	d		Each of the 3 vertices in K_3 must be connected to each of the 4 vertices in K_4. $3 \times 4 = 12$	B1(A02.1) B1(A01.1) B1(A01.1) [10]	Appropriate written explanation K_3 has 3 vertices, K_4 has 4 vertices (or K_n has n vertices) Appropriate calculation shown (leading to the value 12, as given)	K_7 has 21 arcs K_3 has 3 arcs and K_4 has 6 arcs $21 - (3 + 6) = 12$																								

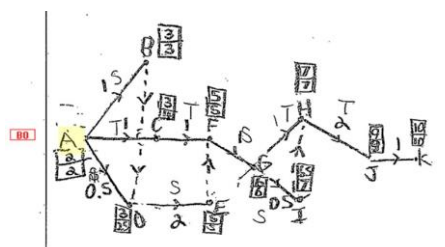
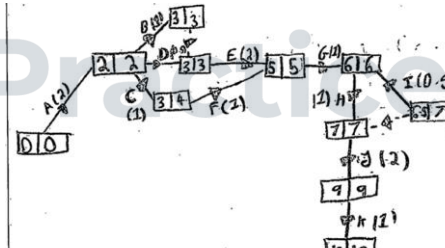


Question			Answer/Indicative content	Marks	Guidance	
			Total	10		
18	a		$^{15}C_2 = 105$	M1(A01.1) A1(A01.1)	$^{15}C_2$ or equivalent seen or implied 105 as final answer	If M0 scored then SC1 for 210 as final answer
	b		Some numbers can be made in more than one way, e.g. 1 and 11 give 111 both ways round	B1(A02.4)	Any suitable example that shows two different orders that give the same result.	e.g 1 followed by 13 or 11 followed by 3 both give 113
	c		Lead digit is 1 when first card is 1 or any of 10, 11, 12, 13, 14, 15.	B1(A02.4)	Identifying that double digit cards also start with 1 7 of the cards have lead digit 1	Benford's law BOD 6 or 8
	d		He does not need to actually find these primes, only find out how many there are.	B1(A02.4) [5]	It is an enumeration problem	Constructing some solutions would not necessarily give a full count
			Total	5		



Question			Answer/Indicative content	Marks	Guidance																																																																									
19		a	 Forward pass Minimum project completion time = 10 days Backward pass Critical activities: A, B, E, G, H, J, K <table border="1" data-bbox="300 1364 756 1552"><thead><tr><th>Activity</th><th>C</th><th>D</th><th>F</th><th>I</th></tr></thead><tbody><tr><td>Float (days)</td><td>1</td><td>0.5</td><td>1</td><td>0.5</td></tr></tbody></table>	Activity	C	D	F	I	Float (days)	1	0.5	1	0.5	B1(AO3.3)	Activity network attempted	For reference: durations <table border="1" data-bbox="1257 304 1476 459"><thead><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th></tr></thead><tbody><tr><td>2</td><td>1</td><td>1</td><td>0.5</td><td>2</td><td>1</td></tr></tbody></table> <table border="1" data-bbox="1257 490 1487 645"><thead><tr><th>G</th><th>H</th><th>I</th><th>J</th><th>K</th></tr></thead><tbody><tr><td>1</td><td>1</td><td>0.5</td><td>2</td><td>1</td></tr></tbody></table> <table border="1" data-bbox="1257 743 1489 976"><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr></thead><tbody><tr><td>ES</td><td>0</td><td>2</td><td>2</td><td>2</td><td>3</td></tr><tr><td>LF</td><td>2</td><td>3</td><td>4</td><td>3</td><td>5</td></tr></tbody></table> <table border="1" data-bbox="1257 1010 1487 1240"><thead><tr><th>F</th><th>G</th><th>H</th><th>I</th><th>J</th><th>K</th></tr></thead><tbody><tr><td>3</td><td>5</td><td>6</td><td>6</td><td>7</td><td>9</td></tr><tr><td>5</td><td>6</td><td>7</td><td>7</td><td>9</td><td>10</td></tr></tbody></table>	A	B	C	D	E	F	2	1	1	0.5	2	1	G	H	I	J	K	1	1	0.5	2	1		A	B	C	D	E	ES	0	2	2	2	3	LF	2	3	4	3	5	F	G	H	I	J	K	3	5	6	6	7	9	5	6	7	7	9	10	M1(AO1.1)	At least one merge correct		
Activity	C	D	F	I																																																																										
Float (days)	1	0.5	1	0.5																																																																										
A	B	C	D	E	F																																																																									
2	1	1	0.5	2	1																																																																									
G	H	I	J	K																																																																										
1	1	0.5	2	1																																																																										
	A	B	C	D	E																																																																									
ES	0	2	2	2	3																																																																									
LF	2	3	4	3	5																																																																									
F	G	H	I	J	K																																																																									
3	5	6	6	7	9																																																																									
5	6	7	7	9	10																																																																									
	A1(AO3.4)	10																																																																												
	M1(AO1.1)	At least one burst correct																																																																												
	A1(AO3.4)	A, B, E, G, H, J, K																																																																												
	M1(AO1.1)	Any one float correct																																																																												
	A1(AO3.4)	All correct																																																																												
	[7]	<u>Examiner's Comments</u>																																																																												
		The specification requires activity-on-arc. Candidates coped well with this, even though for some centres this will have been a new topic. In this particular case, examiners were tolerant of missing directions and missing or extra dummy activities, but this will not necessarily be																																																																												



Question			Answer/Indicative content	Marks	Guidance																																								
					the case on future papers. Exemplar 9  Minimum project completion time = 10 mins Critical activities: A, B, E, G, H, J, K <table><tr><td>Activity</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td></tr><tr><td>Float (days)</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td><td>0.5</td></tr></table> This candidate has used activity on node so gets 0 for the activity network, however the forward pass leads to the correct minimum project completion time (although the units are claimed as mins instead of days) and the backward pass leads to the correct critical activities. Activity C is missing from the list of floats. Exemplar 10  Minimum project completion time = 10 Critical activities: A, B, E, G, H, J, K <table><tr><td>Activity</td><td>C</td><td>D</td><td>F</td><td>I</td><td></td><td></td><td></td></tr><tr><td>Float (days)</td><td>1</td><td>0.5</td><td>1</td><td>0.5</td><td></td><td></td><td></td></tr></table> This is a fully correct solution.	Activity	A	B	C	D	E	F	G	H	I	J	K	Float (days)	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	Activity	C	D	F	I				Float (days)	1	0.5	1	0.5			
Activity	A	B	C	D	E	F	G	H	I	J	K																																		
Float (days)	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5																																		
Activity	C	D	F	I																																									
Float (days)	1	0.5	1	0.5																																									



Question			Answer/Indicative content	Marks	Guidance	
		b	e.g. Activities may not start on time because of delays May need to wait for specialist equipment or other people.	B1(A03.5b) B1(A03.5b) [2]	May give specific examples Delays or activities overrunning May want to avoid splitting an activity overnight when other people are involved	Any issue relating to activity start / finish times / durations Any resourcing issue (including availability of Sheona and Tim) <u>Examiner's Comments</u> Most candidates were able to give either a reason why the durations may be extended or a resourcing issue that would affect the order in which the activities can be carried out. Some candidates were able to give one of each.
		c	Activity E cannot start until Sheona has completed A, B and D. These take $2+1+0.5 = 3.5$ days, so E, cannot start until 3.5 days have elapsed. But E is a critical activity, (so delaying E will delay the entire project).	E1(A02.4) E1(A01.1) [2]	A delay in a critical activity will delay the whole project.	E.g. identifying that B and D cannot be done at the same time. This affects critical activity E (or G) <u>Examiner's Comments</u> Usually candidates realised that two people cannot do activities B, C and D at the same time, or more specifically that Sheona cannot do B and D at the same time. This will delay the start of activity E, which is a critical activity, and hence delay the whole project.



Question			Answer/Indicative content	Marks	Guidance	
		d	(i) Each is occupied for a total of 8 days, So the longest rest either can have is 6 days	M1(A01.1) A1(A01.1) [2]	14 – 6 or (either) busy for 8 6 Allow if the rest for either S or T is identified as 6 days	May be implied from Answer 6 (S does ABDEGI rests for 6 days and then does K, T does ACF rests for 6 days and then does HJK)
					<u>Examiner's Comments</u> Part (d) tested candidates' understanding of (total) float. Each person is busy for 8 of the 14 days. Sheona can rest for 6 days between I and K, Tim can rest for 6 days between F and H.	
			(ii) Activity G must start at the earliest on the afternoon of on day 6 and at the latest on the morning of day 10. Sheona is busy for 6.5 days before the end of activity G (and 1.5 days after activity G) and Tim is busy for 4 days after the end of activity G (and 4 days before the start of activity G). If activities A to I are done as early as possible and activities J, K as late as possible then Sheona and Tim are both on a break from the afternoon of day 8 to the end of day 11 = 3.5 days	M1(A01.1) A1(A01.1) [2]	Using the timing of activity G Or using minimum completion time = 10.5 days Or equivalent reasoning about early/late times 3.5	May be implied from Answer 6
					<u>Examiner's Comments</u> Some candidates identified the longest break that Sheona and Tim can take together. The problem was sorting out when activities G, H and I should happen.	



Question			Answer/Indicative content	Marks	Guidance													
			(iii) A, E, G, H, J, K = 3.5 days <table border="1"><tr><td>B</td><td>C</td><td>D</td><td>F</td><td>I</td></tr><tr><td>4</td><td>5</td><td>4.5</td><td>5</td><td>4</td></tr></table>	B	C	D	F	I	4	5	4.5	5	4	M1(A01.1) A1(A01.1) [2]	Some candidates realised that the only ‘clash’ was between B and D so Sheona and Tim could complete the project in 10.5 days and then take a 3.5 day break. <table border="1"><tr><td>Activities with float = 3.5 days A, E, G, H, J, K</td><td rowspan="2">Accept with 1 or 2 errors in total</td></tr><tr><td>Others correct</td></tr></table> <u>Examiner’s Comments</u> Most of the activities, taken in isolation, had 3.5 days of float. The activities that were previously non-critical had longer floats. The float on B and D depended on whether B came before or after D. Whilst the application of scheduling and cascade charts would be a valid approach for the latter parts of question 6, this question does not require candidates to construct either a schedule or a cascade chart, which is beyond the AS content of the full A Level specification.	Activities with float = 3.5 days A, E, G, H, J, K	Accept with 1 or 2 errors in total	Others correct
B	C	D	F	I														
4	5	4.5	5	4														
Activities with float = 3.5 days A, E, G, H, J, K	Accept with 1 or 2 errors in total																	
Others correct																		
			Total	17														



Question			Answer/Indicative content	Marks	Guidance	
20		a	One luxury pack would not give enough cards with flowers. One standard or economy would not give enough handmade cards.	B1(A02. 2a) B1(A02. 2a) [2]	Explaining why one luxury pack is not enough Explaining why one standard pack or one economy pack is not enough No pack has 8 handmade cards and 8 cards with flowers	One luxury pack only has 20 cards (Alice needs 25 cards) One standard pack only has 5 handmade cards and an economy pack has none (Alice needs 8) <u>Examiner's Comments</u> Candidates needed to show how just one pack of each of three different types did not fulfil the requirements. One pack of luxury cards only has 5 cards with flowers when 8 are needed, or only has 20 cards in total when 25 are needed. One pack of standard cards or one pack of economy cards do not have 8 handmade cards. Some candidates just repeated the wording from the question and quite a few misread the number of cards with animals as 8 instead of 4.

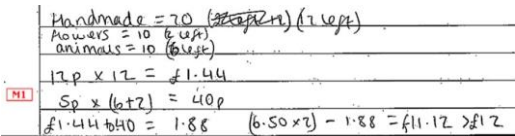
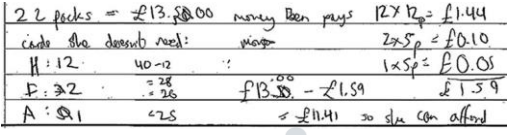
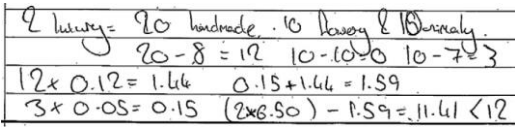
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Question			Answer/Indicative content	Marks	Guidance	
		b	(i) 1 luxury and 1 standard 1 luxury and 1 economy 2 standard (ii) 2 standard	M1(A01.1) A1(A01.1) [2] B1(A01.1) [1]	Any one of these All three and no others (or any extras have been rejected) <u>Examiner's Comments</u> Many correct solutions. Some with one combination missing or an extra, incorrect, combination (such as EEE). <u>Examiner's Comments</u> Candidates usually identified SS as the cheapest, although some claimed LE.	In any form, e.g. LS or SL
		c	Two luxury packs satisfies all the requirements but costs £13. However Ben will buy back 15 cards. Ben can buy back up to 12 handmade cards $(12 \times 12p) + (3 \times 5p) = £1.59$ Giving net cost £11.41 Or a valid calculation, with Ben buying 15 cards, with explanation of how many of each type he bought.	M1(A01.1) A1(A01.1) [2]	Description or calculation showing which surplus cards Ben buys back. e.g. calc leading to one of 11.48, 11.55, 11.62, 11.69. 11.76 (in any form) <u>Examiner's Comments</u> Most candidates realised that Alice could sell (up to) 12 handmade cards to Ben. Some candidates showed that if she did this she would end up with a net expenditure of £11.56. A few candidates realised that Alice only needed to sell Ben 9 handmade cards to end up with a net expenditure that is less than £12.00.	e.g. £13 and a multiple of 12p or a multiple of 5p or a combination of both (shown) e.g. calc leading to one of 11.56, 11.68, 11.80, 11.92 (in any form)



Question	Answer/Indicative content	Marks	Guidance
			Many candidates tried to find the minimum net expenditure for Alice. This was £11.41 by selling Ben 12 handmade cards and 3 cards with flowers or animals. However, many candidates forgot that Alice needed 25 cards so that the maximum that Ben could buy was 15 cards; this usually led to an incorrect answer where Ben bought 12 handmade cards and 8 cards with flowers or animals. Exemplar 6  In this solution Ben buys back 12 handmade cards, 2 cards with flowers and 6 cards with animals. Alice ends up with only 20 cards. Exemplar 7  In this solution Ben buys back 12 handmade cards, 2 cards with flowers and 1 card with animals. £11.41 is the minimum net expenditure for Alice. Exemplar 8  This candidate has used all three types of packs correctly and has simplified the expressions obtained.



Question			Answer/Indicative content	Marks	Guidance	
		d	Handmade: $10x + 5y$ Flowers: $5x + 10y + 10(2 - x - y) = 20 - 5x$ Animals: $5x + 5y + 10(2 - x - y) = 20 - 5x - 5y$ Other: $10y + 20(2 - x - y) = 40 - 20x - 10y$	B1(A01.1) M1(A03.1b) A1(A01.1) A1(A01.1) [4]	Correct expression for handmade Using $2 - x - y$ appropriately Correct expression for one of flowers, animals, other All of these three correct, simplified <u>Examiner's Comments</u> Candidates usually had the correct expression for the number of handmade cards, but often forgot to include the economy packs when calculating the other numbers of cards. The total number of packs is 2 so the number of economy packs is $(2 - x - y)$.	Or flowers = $5x + 10(2 - x)$
		e	$650x + 500y + 400(2 - x - y)$ $-12(10x + 5y - 8) - 5(10x + 25y + 40(2 - x - y) - 17)$ $= 581 + 280x + 115y$	M1(A01.1) M1(A03.1b) A1(A01.1) [3]	$250x + 100y + 800$ $30x + 15y - 219$ cao <u>Examiner's Comments</u> Most candidates were able to find an expression for the cost of buying the packs and subtract an expression for the amount that Ben pays. To achieve the method marks. However, the economy packs were often left out and, even when all three types of pack were used, candidates often had Ben buying 20 cards instead of 15.	Accept $650x + 500y$ Accept $-12(\dots) - 5(\dots)$

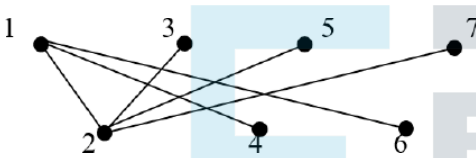


Question			Answer/Indicative content	Marks	Guidance															
		f	Min when x is as small as possible and then y is as small as possible whilst still satisfying Alice's requirements $x = 0$ and $y = 2$ £8.11	M1(AO1.1) A1(AO1.1) [2]	<table border="1"><tr><td>x</td><td>y</td><td>$581 + 280x + 115y$</td></tr><tr><td>0</td><td>2</td><td>811</td></tr><tr><td>1</td><td>0</td><td>861</td></tr><tr><td>1</td><td>1</td><td>976</td></tr><tr><td>2</td><td>0</td><td>1141</td></tr></table> Using their objective Or a fresh start cao <u>Examiner's Comments</u> Some candidates made a fresh start and calculated Alice's net expenditure to buy SS, others used their expression from part (e) to calculate the cost of a valid combination. Quite a few set $x = 0$ and $y = 0$ to minimise the cost, but this corresponded to EE which has no handmade cards.	x	y	$581 + 280x + 115y$	0	2	811	1	0	861	1	1	976	2	0	1141
x	y	$581 + 280x + 115y$																		
0	2	811																		
1	0	861																		
1	1	976																		
2	0	1141																		
			Total	16																

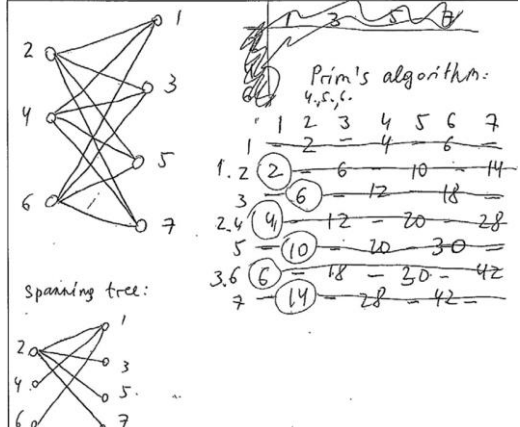


Question			Answer/Indicative content	Marks	Guidance	
21		a	12	B1(A01.2) [1]	cao	<u>Examiner's Comments</u> This was just a test of specification item 7.02e, knowing that $K_{m,n}$ has mn arcs. The answer was 12.
		b	Must start at one of 1, 3, 5, 7 4 choices $\times$ 3 choices $\times$ 3 choices $\times$ 2 choices $\times$ 2 choices = 144 But each path can be travelled in either direction So $144 \div 2 = 72$	B1(A01.1) M1(A02.5) A1(A02.4) [3]	May be implied from working Using multiplicative principle $4! \times 3! (=144)$ 72, from correct working seen <u>Examiner's Comments</u> Many candidates realised that the path needed to 'zig-zag' between the two sets, starting at one of the four vertices {1, 3, 5, 7}. Quite a few candidates achieved the value 144 (or 36 if the initial choice of a value from four was overlooked). The direction of travel does not matter so each path has been counted twice and the total number of paths is 72.	Allow $3 \times 3 \times 2 \times 2 \times 1 \times 1 (=36)$ $\Rightarrow$ B1 M1 144 with no working or incorrect working $\Rightarrow$ B0 M1



Question			Answer/Indicative content	Marks	Guidance
		c	(i) e.g. Using Kruskal's algorithm $1 \times 2 = 2$ $1 \times 4 = 4$ $1 \times 6 = 6$ and $2 \times 3 = 6$ $2 \times 5 = 10$ $3 \times 4 = 12$ forms a cycle $2 \times 7 = 14$... 	B1(A01.1) B1(A01.1) B1(A01.1)	Arc weights correct [at least six correct arc weights shown in a list or on a diagram] Evidence of using Kruskal or Prim [K: need not list all arcs] [P: may start at any vertex] These six arcs chosen (list or tree) <u>Examiner's Comments</u> Some candidates used Kruskal's algorithm and some used Prim's algorithm (often in tabular/matrix form). Candidates needed to at least state the algorithm being used or show sufficient working to make the method obvious. A few candidates relabelled the vertices as A, B, C,... which was accepted provided a key was provided. Some candidates forgot that the arcs always connect an odd number to an even number.



Question			Answer/Indicative content	Marks	Guidance
			(ii) 42	B1(A01.1) [2]	Exemplar 5  This candidate has shown the arc weights in the table, has used Prim's algorithm and has also drawn the minimum spanning tree. cao <u>Examiner's Comments</u> The MST had weight 42.
			Total	8	

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Question			Answer/Indicative content	Marks	Guidance																					
22		i	<table border="1"><thead><tr><th></th><th>P</th><th>Q</th><th>R</th></tr></thead><tbody><tr><td>X</td><td>(1, 4)</td><td>(5, 3)</td><td>(2, 6)</td></tr><tr><td>Y</td><td>(5, 2)</td><td>(1, 3)</td><td>(0, 1)</td></tr><tr><td>Z</td><td>(4, 3)</td><td>(3, 1)</td><td>(2, 1)</td></tr><tr><td>Worst for cols</td><td>2</td><td>1</td><td>1</td></tr></tbody></table>		P	Q	R	X	(1, 4)	(5, 3)	(2, 6)	Y	(5, 2)	(1, 3)	(0, 1)	Z	(4, 3)	(3, 1)	(2, 1)	Worst for cols	2	1	1	M1(A03.1a)	Using worst pay-off in column(s) (2, 1, 1)	Worst result in column P (for player on columns) is 2
				P	Q	R																				
			X	(1, 4)	(5, 3)	(2, 6)																				
Y	(5, 2)	(1, 3)	(0, 1)																							
Z	(4, 3)	(3, 1)	(2, 1)																							
Worst for cols	2	1	1																							
2 > 1 so P is better than Q and R	A1(A03.2a)	Maximin (in column P)	Explaining why P (as given)																							
					<u>Examiner's Comments</u>																					
					[2]																					
					There is no need to convert the table in any way. The expectation was that candidates would consider the worst outcome for each column and hence show that column P has the largest value, ie the best worst outcome for P. Some candidates tried to make a zero-sum game, for example by subtracting the scores in each cell, or assumed that the pay-offs for the player on columns were the negatives of either r or c (for example finding the column minimax).																					
					Those who used the pay-offs for the player on columns were usually successful.																					
					Exemplar 4																					
					<table border="1"><thead><tr><th></th><th>P</th><th>Q</th><th>R</th></tr></thead><tbody><tr><td>X</td><td>(1, 4)</td><td>(5, 3)</td><td>(2, 6)</td></tr><tr><td>Y</td><td>(5, 2)</td><td>(1, 3)</td><td>(0, 1)</td></tr><tr><td>Z</td><td>(4, 3)</td><td>(3, 1)</td><td>(2, 1)</td></tr></tbody></table> Worst outcome (worst pay-off) $\uparrow$ 2 2 is highest - lowest pay-off (Maximin) therefore P is the columns play safe strategy			P	Q	R	X	(1, 4)	(5, 3)	(2, 6)	Y	(5, 2)	(1, 3)	(0, 1)	Z	(4, 3)	(3, 1)	(2, 1)				
	P	Q	R																							
X	(1, 4)	(5, 3)	(2, 6)																							
Y	(5, 2)	(1, 3)	(0, 1)																							
Z	(4, 3)	(3, 1)	(2, 1)																							
					This candidate has shown the 'worst outcome' in each column and explained that column P is the maximin.																					

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Question			Answer/Indicative content	Marks	Guidance	
		ii	e.g. If player on columns plays safe (P) then player on rows does best by playing Y (1, 5, 4). But then player on columns does better by changing from P to Q.	M1(A02.1) A1(A02.2a) [2]	Play-safe for player on rows is Z (1, 0, 2) (Y, P) has a better pay-off for the player on rows than (Z, P) Or cell (P, Z) $\neq$ (2, 2) i.e. the play-safe values <u>Examiner's Comments</u> Several candidates assumed that the game was zero-sum and discussed the values of what they thought were the play-safe strategies. The play-safe for rows was strategy Z, where, for the player on rows, the row minima (1, 0, 2) are maximised. However, cell (Y, P) gives the player on rows a better pay-off (of 5) than cell (Z, P).	
		iii	$A = \{y : y \leq 4\}$	B1(A01.1) [1]	Or equivalent Allow, e.g. $A \leq 4$ Allow $y < 4$ or $y \leq 3$ But not just a single specific value <u>Examiner's Comments</u> The set A was presented in various ways. Some candidates assumed that the value of y had to be non-negative and some assumed that the value of y needed to be an integer. Any reasonable form was allowed. Because of strong/weak dominance examiners accepted both $A = \{y : y < 4\}$ and $A = \{y : y \leq 4\}$.	

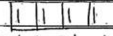
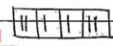
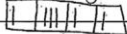


Question			Answer/Indicative content	Marks	Guidance
		iv	In row Z, column P has the best pay-off for the player on columns	B1(A02.4) [1]	Row Z: $3 > 1$ (for player on cols) For the player on cols $P > Q$ in row X (or Z) and $P > R$ in row Z <u>Examiner's Comments</u> Some candidates showed that P dominates one of Q or R but did not show that P was not itself dominated by the other. Many candidates were confused about whether they should be using the pay-offs for the player on rows or the player on columns. Some gave apparently correct comparisons but did not explain which row and columns they were comparing. Some candidates said that P cannot be redundant because it is the play-safe strategy, but the value of p is no longer necessarily 2. If the player on rows plays strategy Z, the player on columns does better by playing P than by playing either Q or R, since $3 > 1$, so P cannot be dominated by either Q or R, and hence is not redundant.
			Total	6	

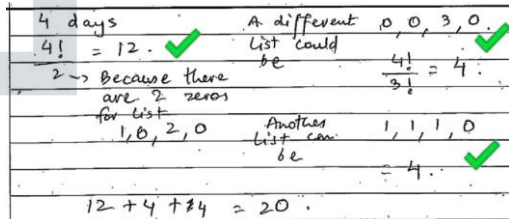


Question			Answer/Indicative content	Marks	Guidance
23		a	There must be at least one day on which Mo eats at least two doughnuts	B1(A01.1) [1]	Or equivalent <u>Examiner's Comments</u> Many excellent answers, saying that Mo must eat at least two doughnuts on at least one day. Some candidates made statements about the number of doughnuts eaten that were not consequences of the pigeonhole principle (such as stating the average number of doughnuts eaten per day).
		b	The maximum on any one day is 3 any day next to the 3 must be a 1, $3+1 = 4$ Or 2 on two days and 1 on two days If the 2's are on adjacent days this gives 4 If not there is a 2 on (at least one) end so the other three days total 4	E1(A02.1) M1(A01.1) A1(A03.1b) [3]	1 1 <u>1 3</u>, 1 <u>1 3</u> 1, <u>1 3</u> 1 1, <u>3 1</u> 1, <u>3 1</u> 1 1 Start to consider cases for 2 2 1 1 May be implied from working May list and identify where total = 4 Dealing with both 2 2 1 1 situations and describing or showing 4's <u>Examiner's Comments</u> Candidates needed to discuss all the various options. Some candidates described the different possibilities and explained how each fitted the requirements. Several candidates listed the various possibilities and then, for example, underlined the days when there were exactly 4 doughnuts. 2 2 1 1, 1 2 2 1 or 1 1 2 2 2 1 1 2, 1 2 1 2 or 2 1 2 1



Question			Answer/Indicative content	Marks	Guidance
					Exemplar 2  pigeonhole = days (4). Eats at least one a day.  The other 2 doughnuts can go in any.  If at end either end, it will take 3 consecutive days. If remaining 2 in any day, it will take 2 consecutive days.  Those are the only ways, so it must take either 2 or 3 days for him to eat 4. This candidate has only given one example of each type. This is not sufficient to get any marks.

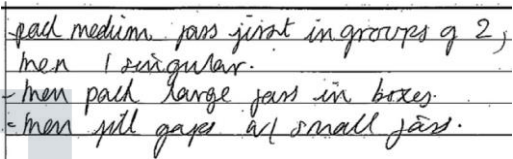


Question			Answer/Indicative content	Marks	Guidance
		c	$4! \div 2! = 12$ arrangements of 0, 0, 1, 2 4 arrangements of 0, 1, 1, 1 4 arrangements of 0, 0, 0, 3	B1(A02.4) B1(A01.1) B1(A01.1) [3]	Or ${}^4C_2 \times 2 = 12$ or write out these 12 Or write out these 4 Or write out these 4 <u>Examiner's Comments</u> Often candidates tried to record all the possibilities, many of these were successful but inevitably sometimes some were omitted or repeated. Some candidates gave calculations that led to the given value 20 but with no justification for why these were a valid way to count the possibilities. Exemplar 3  This candidate has considered the three types of list and then counted how many permutations there are for each.
			Total	7	



Question			Answer/Indicative content	Marks	Guidance	
24		a	Crate 1: 12 small jars Crate 2: 5 small and 1 medium jar Crates 3 and 4: 3 medium jars each Crates 5, 6 and 7: 1 large jar each	M1(A01.1) A1(A01.1) [2]	Crates 1 and 2 correct All correct, in this order <u>Examiner's Comments</u> This part was answered successfully by most of the candidates. Some candidates used a coding, such as 1 = S, 4 = M, 9 = L, this was fine provided a key was given.	May use S, M, L provided intention is obvious If using, e.g. 4S for M, must have defined this for A1
		b	E.g. Crates 1, 2 and 3: 1 large jar and 3 small jars each Crates 4 and 5: 3 medium jars each Crate 6: 1 medium and 8 small jars There is no spare capacity/six full crates /all full	M1(A03.1b) A1(A01.1) B1(A02.4) [3]	Attempt a packing using 6 crates o.e. Pack 17 small, 7 medium and 3 large in 6 boxes, in any order $(17 + 7 \times 4 + 3 \times 9) \div 12 = 72 \div 12 = 6$ <u>Examiner's Comments</u> The most successful answers were those that listed the packing in six crates and then either stated that all the jars had been packed and all the crates were full or calculated the minimum number of crates as $72 \div 12 = 6$. A few candidates left out some of the jars.	For reference: Crate size = 12 Medium = $\times 4$, large = $\times 9$



Question			Answer/Indicative content	Marks	Guidance
		c	e.g. Put each large jar in a box, then put two medium jars in boxes until no more pairs are possible. Put any remaining medium jar in another box and fill in the spaces with small jars.	E1(A03.1b) [1]	Description of a general minimising strategy that has at most 2 medium jars in any box <u>Examiner's Comments</u> Candidates needed to use a strategy that would both minimise the number of boxes and ensure that no box had more than 2 medium jars. Exemplar 1  This candidate packs pairs of medium jars, putting the remaining medium jar and the large jars into separate boxes, and then fills up the spaces with small jars.
			Total	6	



Question			Answer/Indicative content	Marks	Guidance	
25		a	(i) Total is at least 12 pages but editorial uses 0.5 pages (ii) $x \geq 3$ so $x > 0$ $z \geq$ a quarter of total $\geq 0.25(12)$ so $z > 0$ Only need $y \geq 0$ (iii) $z \geq 0.25(x + y + z + 0.5)$ $\Rightarrow 8z \geq 2x + 2y + 2z + 1 \Rightarrow$ given result	B1(AO 2.1) [1] E1(AO 2.1) [1] M1(AO 1.1) A1(AO 1.1) [2]		
		b	When $z = 4$ the constraints become: $x + y \geq 7.5$ $x \geq 3$ $y \leq 1.5$ $x + y \leq 11.5$ $x + y > 3.5$ $y \geq 0$	M1(AO 1.1) A1(AO 1.1) M1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [5]	Putting $z = 4$ in $x + y + z \geq 11.5$ to get $x + y \geq 7.5$ Putting $z = 4$ in $2x + 2y - 2z + 1 \geq 0$ and $2x + 2y - 6z + 1 \leq 0$ (seen or implied from graph) Graphing line $x + y = 11.5$ Graphing line $x + y = 7.5$ Feasible region correct	



Question			Answer/Indicative content	Marks	Guidance	
		c	(i) Optimum at $x = 10, y = 1.5$ giving $P = 3600$ (ii) $z \geq 0.25 \times \text{total number of pages so } k \geq 3$ When $k = 3$, feasible region is bounded by $0 \leq y \leq 1.5$ and $8.5 \leq x + y \leq 8.5$ So optimum at $x = 7, y = 1.5$, giving $P = 2700$ For $3 \leq k \leq 4$, feasible region is bounded by $0 \leq y \leq 1.5$ and $11.5 - k \leq x + y \leq 3k - 0.5$ So optimum at $x = 3z - 2, y = 1.5$, giving $P = 900k - 600 + 600 = 900k$	M1(AO 1.1) A1(AO 1.1) [2] B1(AO 2.4) B1(AO 1.1) M1(AO 2.4) M1(AO 1.1) A1(AO 1.1) [5]	A vertex of their feasible region 3600 $k \geq 3$ $y = 1.5$ in any case other than $k = 4$ $x + y \leq 3k - 0.5$ General case, optimum at $x = 3k - 2$ $P = 900k$	May use z or k at any point
			Total	16		



Question			Answer/Indicative content	Marks	Guidance															
26		a	25	B1(AO 1.2) [2]	5×5															
		b	5! – 4! 96	M1(AO 1.1) A1(AO 1.1) [2]	4 × 4!	Using 4! appropriately														
		c	Three of the $p(r)$ are odd, so at least one of $p(1), p(3), p(5)$ must be odd. If r is odd and $p(r)$ is odd then $r - p(r)$ is even hence the product is even.	M1(AO 1.1) A1(AO 1.1) A1(AO 1.1) [3]	Evidence of using pigeonhole principle to conclude that at least one of $p(1), p(3), p(5)$ is odd Showing that there must be an even term in the product Deducing given result															
		d	Result is not necessarily true. If $\{p(1), p(3), p(5)\} = \{2, 4, 6\}$ and $\{p(2), p(4), p(6)\} = \{1, 3, 5\}$ then $r - p(r)$ is always odd and the product is odd.	B1(AO 2.4) E1(AO 2.3) [2]	No, with an attempted reason or counterexample e.g. <table border="1"> <tr> <td>r</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td>$p(r)$</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>1</td></tr> </table>	r	1	2	3	4	5	6	$p(r)$	2	3	4	5	6	1	$p(r)$ odd if p even and $p(r)$ even if r odd $\Rightarrow r - p(r)$ always odd
r	1	2	3	4	5	6														
$p(r)$	2	3	4	5	6	1														
			Total	8																



Question			Answer/Indicative content	Marks	Guidance	
27		a		M1(AO 3.3) A1(AO 3.3) A1(AO 3.3) [3]	Need not show durations A, B, C, D and E correct F and G correct H correct and no extra dummies	
		b	Minimum completion time = 14 (hours)	M1(AO 3.4) A1(AO 1.1) [2]	Use of a forward pass (may be implied from minimum time = 14) 14	Working may be seen on answer to (a)
		c	(i) A is critical so this will increase the time to 14.5 hours (ii) D has 1 hour of float, so this will increase the time to 15 hours (iii) G becomes critical instead of F, time becomes 13	B1(AO 3.5a) [1] B1(AO 3.5a) [1] B1(AO 1.1) [1]	14.5 15 13	
		d	e.g. extend duration of H to include the break	B1(AO 3.5c) [1]		
			Total	9		



Question			Answer/Indicative content	Marks	Guidance																				
28		a	Sum of points scored for each combination of strategies is 8. Subtract 4 from each score to give the following for Lee: <table><tr><td></td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>P</td><td>1</td><td>4</td><td>-1</td><td>0</td></tr><tr><td>Q</td><td>0</td><td>-2</td><td>3</td><td>1</td></tr><tr><td>R</td><td>-2</td><td>-3</td><td>1</td><td>-1</td></tr></table> (New score for Maria = - new score for Lee)		W	X	Y	Z	P	1	4	-1	0	Q	0	-2	3	1	R	-2	-3	1	-1	M1(AO 1.1a) A1(AO 1.1) [2]	May be implied from a correct zero-sum pay-off matrix. Correct zero-sum pay-off matrix for either player. (Each) 'sum = 8', Or 'score - 4'.
	W	X	Y	Z																					
P	1	4	-1	0																					
Q	0	-2	3	1																					
R	-2	-3	1	-1																					

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Question			Answer/Indicative content	Marks	Guidance																					
		b	R is dominated by Q Let Lee choose randomly between P and Q, choosing P with probability p (or $P(Q) = q$) <table><tr><th>If Maria chooses</th><th>Lee expects</th><th>or</th><th>or</th></tr><tr><td>W</td><td>p</td><td>$1 - q$</td><td>p</td></tr><tr><td>X</td><td>$6p - 2$</td><td>$4 - 6q$</td><td>$4p - 2q$</td></tr><tr><td>Y</td><td>$3 - 4p$</td><td>$4q - 1$</td><td>$-p + 3q$</td></tr><tr><td>Z</td><td>$1 - p$</td><td>q</td><td>q</td></tr></table> $W = Z$ when $p = 1 - p$ $p = 0.5$ Choose randomly between P and Q	If Maria chooses	Lee expects	or	or	W	p	$1 - q$	p	X	$6p - 2$	$4 - 6q$	$4p - 2q$	Y	$3 - 4p$	$4q - 1$	$-p + 3q$	Z	$1 - p$	q	q	B1(AO 1.1) M1(AO 3.1a) A1(AO 3.2a) M1(AO 3.1a) A1(AO 1.1) M1(AO 3.1a) A1(AO 3.2a) B1(AO 3.2a) [7]	$-2 < 0$, $-3 < -2$, $1 < 3$ and $-1 < 1$ so delete row R Assigning a probability to choosing P and / or Q and calculating four expected returns for Lee (may be implied from correct graph) Correct expected returns in algebraic form A correct graph (if using q reflect in prob = 0.5, if using original values translate vertically by 4). (Their) equations at (their) optimum Probability 0.5 from correct algebraic working Correct interpretation in context	Or SC1 for identifying optimum from graph if no equations seen From $p = 0.5$ from equations or graph Or SC1 for identifying
If Maria chooses	Lee expects	or	or																							
W	p	$1 - q$	p																							
X	$6p - 2$	$4 - 6q$	$4p - 2q$																							
Y	$3 - 4p$	$4q - 1$	$-p + 3q$																							
Z	$1 - p$	q	q																							
			Total	9																						



Question			Answer/Indicative content	Marks	Guidance	
29		a	(i) $BE = 7$ (ii) $CD \geq 2$	B1(AO 1.1) [1] B1(AO 2 2a.) [1]	7	Accept $CD > 2$ but not $CD = 2$
		b	(i) $CE \geq 6$ (ii) $E \begin{smallmatrix} 4 & 9 \\ 12 & 9 \end{smallmatrix} \quad F \begin{smallmatrix} 5 & 12 \\ 13 & 12 \end{smallmatrix}$	B1(AO 2 2a.) [1] B1(AO 1.1) B1(AO 1.1) [2]		Accept $CE > 6$, accept $CE = 6$ 4 and 9 in top boxes at E Labelling at F correct
		c	From F: $12 - 5 = 7 (< 8)$ at D $12 - 3 = 9$ at E so EF From E: $9 - 1 = 8$ so DE From D: $8 - 3 = 5$ at B so BD $8 - CD \leq 6$ so CD included if $CD = 2$ etc. Shortest routes: $A B D E F$ and $A B C D E F$	E1(AO 2.4) B1(AO 1.1) B1(AO 1.1) [3]		Explaining trace back in full at any one vertex (e.g. F) $CD = 2$ or route $A B C D E F$ Route $A B D E F$
			Total	8		



Question			Answer/Indicative content	Marks	Guidance	
30		a	(i) $25! = 25 \times 24 \times \dots \times 2 \times 1$ has 25, 20, 15, 10 and 5 as factors. $25 = 5^2$ and 20, 15, 10, 5 each have a factor of 5, so $25!$ is divisible by 5^6. (ii) $25!$ also has factors 2, $4 = 2^2$ and $8 = 2^3$, for example, so $25!$ is divisible by 2^6 and hence is divisible by 10^6	B1(AO 1.1) M1(AO 1.1) A1(AO 2.4) [3] E1(AO 1.1) [1]	$25!$ Factors 5, 10, etc. $25 = 5^2$ leading to conclusion that $25!$ is divisible by 5^6 $25!$ is also divisible by 2^6	Or pairing factors to make a multiple of 10^6, e.g. $10 \times (2 \times 20 \times 25) \times (4 \times 5 \times 15)$
		b	(i) 28 5 (ii) $144 + 28 + 5 + 1$ 178	B1(AO 1.1) B1(AO 1.1) [2] M1(AO 1.2) A1(AO 1.1) [2]	cao cao Allow for three of these values cao	
		c	$720 - 360 - 144 + 72$ 288	M1(AO 2.1) A1(AO 1.1) [2]	Use of inclusion-exclusion	
			Total	10		



Question			Answer/Indicative content	Marks	Guidance
31		i	Suppose that the kiln capacity is k, then each dish uses up $\frac{k}{15}$ and x dishes use up $\frac{kx}{15}$ (and similarly for mugs and pots), so $\frac{kx}{15} + \frac{ky}{50} + \frac{kz}{200} \leq k \Rightarrow \frac{x}{15} + \frac{y}{50} + \frac{z}{200} \leq 1$ so $40x + 12y + 3z \leq 600$ $a = 40, b = 12, c = 3$ and $d = 600$	M1 M1 A1 [3]	Dealing with any of 15, 50 and 200 appropriately (e.g. $x \div 15$ or $15a$, but not $15x$) An inequality of the form $ax + by + cz \leq d$, with numerical coefficients and $b = 4c$, but not necessarily integers Coefficients 40, 12, 3, 600 or positive integer multiples of these (stated as a, b, c and d or seen in inequality) $40x + 12y + 3z \leq 600$ o.e. (+ integer coefficients) $\Rightarrow$ M1, M1, A1 Examiner's Comments Many candidates offered the inequality as either $x + y + z \leq 265$ or $15x + 50y + 200z \leq \text{some value}$, clearly neither of these are consistent with the upper limits of 15, 50 and 200. Other candidates realised that the coefficients represented the amount of space occupied by one item but were not able to progress beyond that. Those who realised that $d = 15a = 50b = 200c$ (rather than $15x$, etc.) could usually achieve the required inequality.
		ii	x	B1 [1]	cao Examiner's Comments Several candidates realised that the pivot had been the x-column. Some gave z (the next pivot choice rather than the previous one) and a few offered other columns.



Question			Answer/Indicative content	Marks	Guidance																																																						
		iii	[z becomes basic and t becomes non-basic] <table><tr><td>P</td><td>x</td><td>y</td><td>z</td><td>s</td><td>t</td><td>u</td><td>v</td><td>RHS</td></tr><tr><td>1</td><td>0</td><td>2</td><td>0</td><td>0</td><td>3</td><td>0</td><td>-1</td><td>68</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>2</td><td>0</td><td>-4</td><td>32</td></tr><tr><td>0</td><td>0</td><td>2</td><td>1</td><td>0</td><td>4</td><td>0</td><td>-3</td><td>24</td></tr><tr><td>0</td><td>0</td><td>6</td><td>0</td><td>0</td><td>28</td><td>1</td><td>-31</td><td>368</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>-1</td><td>0</td><td>1</td><td>4</td></tr></table>	P	x	y	z	s	t	u	v	RHS	1	0	2	0	0	3	0	-1	68	0	0	1	0	1	2	0	-4	32	0	0	2	1	0	4	0	-3	24	0	0	6	0	0	28	1	-31	368	0	1	0	0	0	-1	0	1	4	B1 M1 M1 A1 [4]	Correct pivot choice (may be implied from resulting tableau) Correct structure with all cells filled in, no blanks 5 different basic columns (0's and a single 1) specifically columns P, x, z (or y) and two of s, t, u RHS values are all non-negative P value in RHS column = 68 [dep on both M marks] Ignore errors in other values provided these conditions are met Examiner's Comments The second iteration was often done well, provided the pivot choice was correct and there were not too many arithmetic errors. Candidates should check that their answer has the correct structure (all cells filled in, basis columns, no negative entries in RHS column).
P	x	y	z	s	t	u	v	RHS																																																			
1	0	2	0	0	3	0	-1	68																																																			
0	0	1	0	1	2	0	-4	32																																																			
0	0	2	1	0	4	0	-3	24																																																			
0	0	6	0	0	28	1	-31	368																																																			
0	1	0	0	0	-1	0	1	4																																																			
		iv	(a) 0 dishes, 0 mugs and 36 pots Profit £72	B1 B1 [2]	36 <u>pots</u> (in context, may imply 0 dishes and 0 mugs) £72, with units Examiner's Comments Interpretation of the given final tableau was poor. Even the candidates who wrote down the numerical values of P, x, y and z correctly did not always interpret the solution in context.																																																						



Question			Answer/Indicative content	Marks	Guidance
		iv	(b) May not be able to sell all the pots Goods may crack in kiln, damaged goods cannot be sold at full price A particular customer may want a dish or a mug Demand for dishes, mugs, pots not constant	B1 [1]	Any of these or similar Need to explain why making e.g. 0 dishes, 0 mugs, 36 pots may not generate the £72 that it should Not discussing slack variables or changing costs (unless it is to take account of e.g. damaged goods) Examiner's Comments The question had already said that the other constraints took account of the amount of clay required, the potter's time and customer demand. The cost of running the kiln will have been incorporated into the profit equation. Most candidates said that to achieve this profit the potter has to sell all the goods, this was fine. Some candidates realised that a pot could be damaged in the kiln and not be suitable for sale at the full price.
			Total	11	



Question			Answer/Indicative content	Marks	Guidance
32		i	$A = (0, 4)$ $B = (0, 2)$ $C = (4, 1)$ $D = (4, 3)$ $E = (3, 5)$	M1 A1 A1 A1 [4]	Any one coordinate correct (0, 4) and (0, 2) both correct (4, 1) and (4, 3) both correct (3, 5) correct Ignore not labelling as A, B, C, D, E or mislabelling Examiner's Comments Most candidates found the coordinates of at least four of the five points of intersection.
		ii	Either checking vertices or using a sliding profit line $B(0, 2): 3(0) + 13(2) = 26$ $C(4, 1): 3(4) + 13(1) = 25$ $[A = 52, D = 51, E = 74]$ Minimum value is 25	M1 A1 [2]	Sight of any $3x + 13y$ for specific numerical values of x, y May be implied from identification of (4, 1) or C or from any of these calculations, may be implied from 25 seen with (4, 1) or from 25 indicated in some way as final answer for min value 25 indicated in some way as final answer for min value (www) Examiner's Comments Either by evaluating P at each of A, B, C, D and E or by using a sliding profit line, candidates were usually able to identify 25 as the minimum feasible value. Some candidates gave the coordinates of the point where this occurred rather than giving the minimum feasible value and a few calculated the value 25 in their list but stated 26 as the minimum.

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Question			Answer/Indicative content	Marks	Guidance
		iii	New vertex of feasible region where $2x + y = 11$ and $x + 4y = 8$ $\Rightarrow \left(5\frac{1}{7}, \frac{5}{7}\right)$ $3\left(5\frac{1}{7}\right) + 13\left(\frac{5}{7}\right) = 24\frac{5}{7}$	M1 A1 A1 [3]	Using <u>these</u> two equations simultaneously (or implied) (5.14, 0.71) (x and y values correct as fractions or to at least 2 dp) 24.7 (correct as a fraction or to at least 3sf) Fractions may be 'top heavy' i.e. $\frac{36}{7}$, $\frac{173}{7}$ Examiner's Comments Usually done well, apart from candidates who were using the wrong pair of simultaneous equations. A few worked out the coordinates of the new vertex but not the value of the objective at that point. Some candidates assumed that the coordinates had to be integer-valued at this stage.

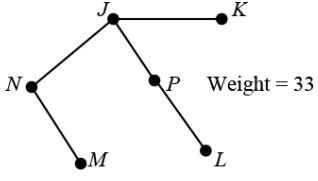


Question			Answer/Indicative content	Marks	Guidance
		iv	(5, 1) is feasible since $2(5) + 1 = 11$ and $5 + 4(1) = 9 > 8$ But $3(5) + 13(1) = 28 > 25$ [or 26]	B1 B1 [2]	Showing that $2x + y \leq 11$ <u>and</u> $x + 4y \geq 8$ at (given point) (5, 1) Numerical evidence (e.g. 11 <u>and</u> 9) not just ticking as checked [Ignore if $-x + 3y \leq 12$ checked] $3x + 13y$ is smaller at (4, 1) [or (0, 2)] than at (5, 1) Sight of 28 (or $15 + 13$ or $3(5) + 13$) <u>and</u> one of 25, 26, C or B 28 and reference to their value from part (ii) or (4, 1) [or (0, 2)] oe (i.e. the candidate has said that the value is from part (ii)) Examiner's Comments There were two things to do here: show that (5, 1) is feasible and show that there are other integer-valued feasible points at which the objective is smaller. To show that (5, 1) is feasible involved showing that the coordinates satisfy the constraints, stating that it is to the left and above the point found in part (iii) is not sufficient. To show that (5, 1) is not the minimum required calculating the value of the objective at (5, 1) and also identifying an integer-valued feasible point with a smaller value of P.
			Total	11	



Question			Answer/Indicative content	Marks	Guidance
33		i	$P - J - N - M - L - K - P$ $6 + 5 + 6 + 11 + 12 + 8 = 48$ miles	M1 A1 [2]	Route starts $P - J - N - M$ 48 Examiner's Comments Most candidates were able to apply the nearest neighbour method correctly, a few returned to P before every vertex had been visited. Nearest neighbour uses the shortest arc from the current vertex to a vertex that has not yet been visited, until all vertices have been visited and the route returns to the start. In this case the route that was obtained by returning to P midway through the route was in fact longer than the nearest neighbour route.
		ii	e.g. $P - L - M - N - J - K - P$ or $P - M - N - J - K - L - P$ = 46 miles = 47 miles	M1 A1 [2]	One of these routes, or in reverse Correct length (46 or 47) for their route (having achieved M1) Examiner's Comments Candidates could usually find one of the two possible improved routes and its weight.



Question			Answer/Indicative content	Marks	Guidance
		iii	$JN = 5$ $JP = 6$ $MN = 6$ $JK = 7$ $KP = 8$ $MP = 8$ $LP = 9$...		M1 At least five of these seven arcs written listed by increasing order of weight, may have MN before JP and / or MP before KP Not just a list of weights and not from diagram A1 Indicating that KP and MP are not chosen B1 Correct tree B1 33 (miles) [4] Examiner's Comments Many correct answers were seen although some candidates did not show that they had considered the arcs in increasing order of weight and some only listed the arcs that had been chosen.
		iv	$k > 9$ LM must be longer than both LP and MN , and the longer of these is $LP = 9$	B1 B1 [2]	k greater than 9 (not $k \geq 10$ and not $k \geq 9$) $LP = 9$ and $MN < 9$, need to explicitly refer to both LP <u>and</u> MN (i.e. least weight arc at each end of new arc) Examiner's Comments Many candidates realised that k had to be greater than 9, but often the explanation assumed that the tree was being built using Kruskal's algorithm and the possibility of a different minimum spanning tree using Prim's algorithm was only occasionally considered. Unless the totals are considered very carefully, if a tree is formed using Prim's algorithm then if M is joined in before L the value of k also needs to be compared with $MN = 6$.



Question			Answer/Indicative content	Marks	Guidance
		v	Shortest closed route that does not use arc LM is $P - M - N - J - K - L - P$ (or in reverse) = 47 miles Shortest route that does use arc LM is $P - L - M - N - J - K - P$ (or in reverse) = $35 + k$ miles So $35 + k < 47 \Rightarrow k < 12$ (hence $9 < k < 12$) For reference	M1 M1 A1 [3]	(Shortest route without LM has length) 47 (soi) (Shortest route using LM has length) $35 + k$ (soi) $k < 12$ (not $k \leq 11$, not $k = 10, 11$ and not $k \leq 12$) [from either M mark] SC1 for $k < 12$ (with or without reference to LK) (or as part of a double inequality, e.g. $9 < k < 12$) if neither M mark has been given Examiner's Comments The answer $9 < k < 12$ was often seen, though $k < 12$ was also accepted. Sometimes candidates went on to assume that the arc weights had to be integer-valued and gave $k = 10, 11$ as their answer. Although the arc weights in the question were all integer-valued this was not stated as a restriction. A small, but not insignificant, number of candidates correctly found the length of the shortest cycle that does not include LM and the length of the shortest cycle that does include LM and used these to deduce that $35 + k < 47$ so $k < 12$.
			Total	13	



Question			Answer/Indicative content	Marks	Guidance
34		i	B and E	B1 [1]	Both correct and no others May list vertex orders but need B and E specifically identified Examiner's Comments Most candidates identified the odd vertices correctly.
		ii	$A - D - F = 13 + 16 = 29$ $A - E - F = 27 + 8 = 35$ Shortest travel time = 29 minutes	M1 A1 [2]	Evidence of considering both $A - D - F$ and $A - E - F$ or sight of <u>both</u> 35 and 29 or mentioning both D and E 29, after considering both possibilities SC1 for 29 without showing evidence of considering alternatives Examiner's Comments Candidates usually calculated the journey times for $A - D - F$ and $A - E - F$ but did not always answer the question asked, which was to find the shortest travel time (29) not to state the route.



updating the temporary label when a node at distance d is shorter than the current best temporary label.



Question			Answer/Indicative content	Marks	Guidance
		iv	Total time for direct routes = 222 then repeat B to $E = 24$ Total time = 246 minutes Repeated arcs BC and CE C is travelled through 4 times	M1 A1 B1 B1 [4]	Attempt at total for <u>half</u> of table i.e. 222 or $444 \div 2$ 246 or 4 hours 6 minutes (cao) [answer 246 o.e. $\Rightarrow$ M1, A1] Accept $B - C - E$ (correct, not ft) but not e.g. '$B - E$ via C' 4 or $3 + 1 = 4$ Examiner's Comments The sum of all the values in the table was given as 444, but each arc appears twice so the sum of the arc weights is 222. Most candidates said that $BE = 24$ needed to be repeated but did not say that this used the arcs BC and CE. This meant that the vertex order of C had to be increased from 6 to 8, but each time the route passes through C it uses up two of these arcs, so C is passed through 4 times and not 8 as several candidates suggested.
			Total	11	



Question			Answer/Indicative content							Marks	Guidance																																																																																				
35		i	<table><tr><th>Line</th><th>N</th><th>P</th><th>M</th><th>A</th><th>B</th><th>Display</th></tr><tr><td>10</td><td>5</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td></td><td>0</td><td>5</td><td></td><td></td><td></td></tr><tr><td>90</td><td></td><td></td><td></td><td>13</td><td></td><td></td></tr><tr><td>100</td><td></td><td></td><td></td><td></td><td>12</td><td></td></tr><tr><td>110</td><td></td><td></td><td></td><td></td><td></td><td>5, 12, 13</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>							Line	N	P	M	A	B	Display	10	5						80		0	5				90				13			100					12		110						5, 12, 13																																												Mark entries in columns for the variables: N, P, M, A, B in the row where they first appear Ignore any row labelled as line 20 or line 60 B1 Line 10, $N = 5$ B1 $P = 0$ and $M = 5$ both on second row M1 $A = 13$ on the next row A1 $B = 12$ on the following row and stopping after (attempted) display Ignore values copied down into lower rows Do not deduce entries from printed display values [4] Examiner's Comments Most candidates were able to trace through the algorithm, although some wrote all their working on one row, instead of using one row for each line when a value was updated. A few made numerical errors, often in calculating the new value of A or B.
			Line	N	P	M	A	B	Display																																																																																						
			10	5																																																																																											
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Question			Answer/Indicative content							Marks	Guidance																																																																													
		ii	<table><tr><th>Line</th><th>N</th><th>P</th><th>M</th><th>A</th><th>B</th><th>Display</th></tr><tr><td>10</td><td>20</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>30</td><td></td><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td>40</td><td></td><td></td><td>20</td><td></td><td></td><td></td></tr><tr><td>50</td><td></td><td></td><td>10</td><td></td><td></td><td></td></tr><tr><td>70</td><td></td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>50</td><td></td><td></td><td>5</td><td></td><td></td><td></td></tr><tr><td>90</td><td></td><td></td><td></td><td>52</td><td></td><td></td></tr><tr><td>100</td><td></td><td></td><td></td><td></td><td>48</td><td></td></tr><tr><td>110</td><td></td><td></td><td></td><td></td><td></td><td>20, 48, 52</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>							Line	N	P	M	A	B	Display	10	20						30		1					40			20				50			10				70		2					50			5				90				52			100					48		110						20, 48, 52									Mark entries in columns for the variables: N, P, M, A, B in the row where they first appear Ignore any row labelled as line 20 or line 60 B1 Line 10, $N = 20$ B1 $P = 1$ with $M = 10$ (second and fourth rows) B1 $P = 2$ with $M = 5$ (fifth and sixth rows) M1 $A = 52$ on the next row A1 $B = 48$ on the following row and stopping after (attempted) display Ignore values copied down into lower rows Do not deduce entries from printed display values [5] Examiner's Comments Many correct answers. Some candidates recorded M changing to 5 in the same line 50 row as where they had recorded M becoming 10, instead of having a new line 50 row following the line 70 row.
Line	N	P	M	A	B	Display																																																																																		
10	20																																																																																							
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110						20, 48, 52																																																																																		

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Question			Answer/Indicative content	Marks	Guidance
		iii	$N = 2^x$ will be repeatedly halved until $M = 1$ and $P = x$ Then $A = 2 \times 2^{x-1} = 2^x$ (or $A = N$) and $B = 2^x - 2^x = 0$ The displayed values (for N, B, A) are $2^x, 0, 2^x$	M1 A1 B1 B1 [4]	Iterate <u>until</u> it = 1, may be from trying a specific case or implied $P = x$, may be deduced from specific cases but need x here $[P = x + 1 \Rightarrow M1, A0] [P = x \Rightarrow M1, A1]$ $A = 2^x$ or $A = N$ (but not $A = 2^P$) $B = 0$ This B1 B1 may be implied from these displayed values in this form and in this order Examiner's Comments Some candidates incorrectly stated that repeatedly dividing by 2 would never give an odd number. Those who realised that M would eventually become 1 (sometimes from having tried out a specific case, such as $N = 8$) often got $P = x$ (although sometimes this value was out by 1) and hence the final values $B = 0$ and $A = 2^x$ (or the original N value).
			Total	13	



Question			Answer/Indicative content	Marks	Guidance
36		i	(a) The nodes are not all of even order.	B1 [1]	Must be written answers. Diagrams may support reasoning but get no marks on their own. It has an odd vertex; it has odd vertices; it has a vertex of order 5 which is odd Need 'odd' (or 'not all even'), not just 'there is a vertex of order 5' Examiner's Comments Some candidates told us everything they knew, or thought they knew, about Josh's graph in this first part. The question had specified that candidates were to answer without doing any calculations and without drawing graphs. Part (a) asked how the given vertex order show that Josh's graph is not Eulerian – Josh's graph has a vertex of odd order (5) so the vertices are not all of even order and the graph cannot be Eulerian.



Question			Answer/Indicative content	Marks	Guidance
		i	(b) Odd nodes occur in pairs so one of the missing values is odd and the other is even, (hence exactly 2 nodes of odd order so semi-Eulerian). OR Total vertex order is even so one more odd node (one odd and one even) (exactly one extra odd) (2 odd nodes altogether)	B1 [1]	Using 'even number of vertices with odd vertex order' to deduce that there are 2 odd vertices (or only one of the missing ones is odd), need not explicitly say 'so semi-Eulerian' Not just stating that there must be 2 odd vertices (or that one of the missing ones is odd) (with or without reference to semi-Eulerian) <u>Not for calculating</u> the missing orders 1, 2 or for using total = 16 Examiner's Comments Part (b) asked how the given vertex orders show that Josh's graph is semi-Eulerian – odd order vertices come in pairs and we know that there is one odd order vertex so one of the missing orders must also be odd (and the other even), which means that there is exactly one pair of odd order vertices and hence the graph is semi-Eulerian. Just stating that a semi-Eulerian graph has two odd order vertices was not enough, candidates needed to explain how they knew that Josh's graph had exactly two odd order vertices.



Question			Answer/Indicative content	Marks	Guidance
		ii	The sum of the vertex orders = $2 \times$ number of arcs = 16 $\Rightarrow$missing orders sum to 3 Simply connected, so vertex order 0 is impossible. Hence missing vertex orders are 1 and 2	M1 A1 [2]	Answer must be labelled as (ii) not credited in (iii) Sum of orders = 16 or twice number of arcs or sight of $2 \times 8 = 16$ or sum of missing orders = 3 or 16 - $(2+2+4+5)$ or $16 - 13$ or use reasoning similar to (iii) using vertex of order 5 Not just from a diagram and not from unsupported answers 1 and 2 Missing orders are 1 and 2 (deduced from a complete and correct reasoned argument including simply connected or $16 - 13$) Examiner's Comments The easiest way to calculate the missing vertex orders was to use the fact that the graph has 8 arcs to deduce that the sum of the vertex orders is 16 and hence the sum of the missing orders is 3. Since the graph is connected the missing orders cannot be 0 and 3, so they must be 1 and 2 (and there are no other possibilities).



Question			Answer/Indicative content	Marks	Guidance
		iii	Given a simply connected graph vertex orders 5, 4, 2, 2, 2, 1 The vertex of order 5 must be joined to <u>each</u> of the others The order 4 is also connected to the three 2's The order 1 is not connected to any of these The order 2's are not connected to each other Hence, all such graphs have the same structure (are isomorphic)	B1 M1 A1 [3]	Do not need to explain why these are the vertex orders Accept other words for 'order' provided intention is obvious Understanding <u>connections</u> for the vertex of order 5 Trying to deal with <u>connections</u> for the other vertices, any one of these three statements or equivalent (e.g. order 4 is connected to three others) Correct argument from a simply connected graph with vertex orders 5,4,2,2,2,1 to only one possible structure Examiner's Comments This last part was asking why all graphs with six vertices of orders 5, 4, 2, 2, 2, 1 must have the same structure (must be isomorphic). Candidates who started from 'it is a simply connected semi-Eulerian graph with six vertices and eight arcs' and tried to deduce that these must be the vertex orders were doomed to fail; there are other possible vertex orders that fit this description (e.g. 4, 3, 3, 2, 2, 2). In general there will be multiple graphs with different structures that have the same set of vertex orders, in this particular case there is only one possible structure, but showing this involves considering which vertices can be connected to which other vertices.
			Total	7	



Question			Answer/Indicative content	Marks	Guidance
37		i	After 1st pass 10 3 6 12 4 5 12 After 2nd pass 3 6 10 4 5 12 12 After 3rd pass 3 6 4 5 10 12 12 After 4th pass 3 4 5 6 10 12 12 After 5th pass 3 4 5 6 10 12 12	M1 A1 B1 [3]	(Lists may be written vertically) First pass has 12 at right-hand end (even if other values are wrong) (first row or first row after original or identified as pass 1) This list after 3rd pass (even if there were earlier errors) List sorted after 4th pass, 5th pass used to check but no 6th pass Examiner's Comments Most candidates applied bubble sort correctly, although some showed every comparison or every swap and then did not indicate which row was the end of each pass. Most candidates worked on horizontal lists although vertical working was also accepted when it was evident that this was what a candidate was doing. A common error was to stop a row too early, when the list could be seen to be sorted but the 'check' pass had not been carried out, others continued to a sixth pass. Bubble sort can be stopped when either there is a pass in which no values are swapped or no more passes are possible.

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Question			Answer/Indicative content	Marks	Guidance
		ii	(a) 40 27 8 14 27 40 8 14 Note: given list was 27 8 14 40, this may appear before the answers	B1 B1 [2]	This list (first or second) This list (first or second) If more than two lists are given, mark the first two only Examiner's Comments Only a few candidates were not able to reverse the process to give at least one of the possible preceding lists. To get the value 27 in the first position after one pass the original list must have started with either 27 40 or with 40 27, in both cases the 40 passes to the end of the list without changing the order of the other three values.
		ii	(b) 2	B1 [1]	'2 more' Ignore any passes shown, B1 is for stating that there are two (more) Examiner's Comments With bubble sort the 'check' pass is part of the sort, so there would be two further passes.
			Total	6	



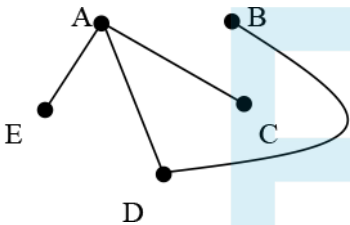
Question			Answer/Indicative content	Marks	Guidance																													
38		a	5 milk chocolate in each type, $4 \times 5 = 20$	B1 (A02.2a) [1]	Using numbers of milk chocolate truffles																													
		b	2 so that there are at least 12 nutty chocolate truffles	B1 (A02.2a) [1]	2 (Assorted boxes) and using numbers of nutty chocolate truffles																													
		c	<table><tr><th colspan="3">Type</th><th>Cost</th></tr><tr><th>Assorted</th><th>No Nuts</th><th>Speciality</th><th>£</th></tr><tr><td>4</td><td>0</td><td>0</td><td>8.00</td></tr><tr><td>3</td><td>1</td><td>0</td><td>7.00</td></tr><tr><td>3</td><td>0</td><td>1</td><td>8.50</td></tr><tr><td>2</td><td>1</td><td>1</td><td>7.50</td></tr><tr><td>2</td><td>0</td><td>2</td><td>9.00</td></tr></table> 3× Assorted and 1× No Nuts	Type			Cost	Assorted	No Nuts	Speciality	£	4	0	0	8.00	3	1	0	7.00	3	0	1	8.50	2	1	1	7.50	2	0	2	9.00	M1 (A01.1) A1(A01.1) B1(A01.1) [3]	Identifying at least three of these five cases for the number of boxes of the different types bought All five cases, and no others, unless explicitly rejected as being infeasible cao	Need not show costs, and if given do not penalise calculation errors May be marked in list
Type			Cost																															
Assorted	No Nuts	Speciality	£																															
4	0	0	8.00																															
3	1	0	7.00																															
3	0	1	8.50																															
2	1	1	7.50																															
2	0	2	9.00																															
		d	(i) P: $8x + 4y \geq 10$ oe W: $7x + 9y \geq 16$ oe $x \geq 0$ and $y \geq 0$ (and integer-valued)	M1 (A03.1b) A1(A01.2) [2]	One of $8x + 4y \geq 10$ and $7x + 9y \geq 16$, in any form Both of these correct, in any form	Allow strict inequalities, but not an equality or with inequality reversed																												



Question			Answer/Indicative content	Marks	Guidance	
	d	(ii)		M1 (A03.3) A1(A01.1)	Plotting lines and identifying feasible region All lines and feasible region correct on scaled axes	For reference: (0, 2.5) (0.59, 1.32) or better (2.28 to 2.29, 0)
	d	(iii)	(0, 3) + 2 small milk + 2 small nutty ⇒ £9.50 (1, 2) + 2 small milk + 2 small nutty ⇒ £8.00 (2, 1) + 2 small milk + 3 small nutty ⇒ £7.00 (3, 0) + 2 small milk + 3 small nutty ⇒ £5.50 (1, 1) + 3 small milk + 3 small nutty ⇒ £6.50 Cheapest solution costs Narendra £5.50	[2] M1 (A03.4) M1(A03.2a) E1(A03.4)	Calculating costs for any integer-valued feasible solution Including sufficient small packs to deal with 'milk' and 'nutty' requirements £5.50	May go straight to solution (from logical reasoning or calculation) E0 if exterior points used as if feasible E0 if correct answer given with insufficient evidence to justify it
			Total	12		

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Question			Answer/Indicative content	Marks	Guidance																																				
39		a	e.g. <table border="1"><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr></thead><tbody><tr><th>A</th><td>-</td><td>9</td><td>5</td><td>4</td><td>2</td></tr><tr><th>B</th><td>9</td><td>-</td><td>7</td><td>5</td><td>7</td></tr><tr><th>C</th><td>5</td><td>7</td><td>-</td><td>6</td><td>8</td></tr><tr><th>D</th><td>4</td><td>5</td><td>6</td><td>-</td><td>5</td></tr><tr><th>E</th><td>2</td><td>7</td><td>8</td><td>5</td><td>-</td></tr></tbody></table> Using Prim's algorithm starting at A AE = 2 AD = 4 AC = 5 (or DB) DB = 5 (or AC) Total weight = 16 		A	B	C	D	E	A	-	9	5	4	2	B	9	-	7	5	7	C	5	7	-	6	8	D	4	5	6	-	5	E	2	7	8	5	-	M1(A01.1a) A1(A01.1) A1(A01.1) B1(A01.1) [4]	Stating which is the starting vertex A valid order of building the tree for their starting vertex, clearly shown (arcs or vertices with arcs indicated on matrix) Correct (labelled) tree Weights need not be shown
	A	B	C	D	E																																				
A	-	9	5	4	2																																				
B	9	-	7	5	7																																				
C	5	7	-	6	8																																				
D	4	5	6	-	5																																				
E	2	7	8	5	-																																				
Copyright © 2024 Exam Papers Practice			b (i) Vertex A b (b) $0 + (4 + 5) + 5 + 4 + 2$ $\Rightarrow 20$	B1 (A01.1) [1]																																					

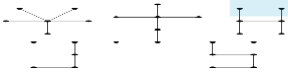


Question			Answer/Indicative content	Marks	Guidance	
		c	$(1 \times 4 - 1) + (2 \times 3 - 1) + (3 \times 2 - 1) + (4 \times 1 - 1)$ $= 3 + 5 + 5 + 3 = 16$	M1(A02.1) M1(A01.1) A1(A01.1) [3]	1×4, 2×3, etc. Subtracting 1 for each pass www	Or by referring to their working in part (a)
		d	Cubic order Approximately $\left(\frac{500}{100}\right)^3 \times 4 = 500$ seconds oe	M1(A01.1) A1(A02.2b) [2]	Stating or using the fact that Prim's has cubic order Must include an indication that this is an approximation	8 minutes 20 seconds
			Total	11		



Question			Answer/Indicative content	Marks	Guidance	
40		a	Total mass is 60, so each bin must hold at least 20kg If it is 20kg then using first-fit: 8 5 7 9 7 1 3 9 3 8 Using first-fit decreasing: 9 9 1 8 8 3 7 7 5 3 But if the capacity was 21kg then the 3 could go into the first bin and the 1 into the second bin, so only three bins are needed So the bins cannot be more than 20kg, so they must be exactly 20kg	B1 (A01.1) M1(A03.1b) A1(A01.1) M1(A03.1b) A1(A01.1) E1(A03.2a) [6]	Consider total mass to find a lower bound for capacity Attempt first-fit with a capacity of at least 20kg All correct, in this order Attempt first-fit decreasing with a capacity of at least 20kg All correct in this order (or a description in words) 20 is also the upper bound, so the capacity is 20	First bin correct, in this order Correct placement of the values 9 9 8 8 7 7
		b	Shuttle sort has quadratic order, as a function of the length of the list, so the time to run first-fit decreasing would be the sum of a quadratic function and a linear function, which is a quadratic function This means that first fit-decreasing has quadratic order in this case	M1(A01.1) A1(A02.2a) [2]	Knowing and using the fact that shuttle sort has quadratic order	$T(n) = O(n^2) + O(n)$ $= O(n^2)$
			Total	8		



Question			Answer/Indicative content	Marks	Guidance	
41		a		B1 (A01.1) [1]	These three graphs and no others	May appear in any equivalent variation
		b	e.g. Six vertices so need 6 positive integers e.g. 5 arcs so sum of vertex orders = 10 e.g. Tree must have at least two 'ends' so at least two vertex orders have value 1	B3 (A01.1 1.1 2.5) [3]	B1 for each independently correct property	Award B2 for three properties if incorrect or inconsistent terminology used
		c	{1, 1, 1, 1, 1, 5} {1, 1, 1, 1, 2, 4} {1, 1, 1, 1, 3, 3} {1, 1, 1, 2, 2, 3} {1, 1, 2, 2, 2, 2}	M1 (A01.1) A1(A01.1) [2]	At least four correct sets All five correct with no extras	
		d		M1 (A01.1) A1(A01.1) [2]	At least four correct graphs Five correct graphs and no extras, unless it is because both versions of {1, 1, 1, 2, 2, 3} are shown	Other possibility for {1, 1, 1, 2, 2, 3} 
			Total	8		



Question			Answer/Indicative content	Marks	Guidance	
42		a	Route e.g. $c - b - a - e - d - c - b - e - d$	B1 (A01.1) [1]	A route that starts and ends at c and d and uses every edge once and only once	
		b	It has exactly two odd vertex orders	B1(A01.2) [1]	Two and only two odd	
		c	In graph 1 there are two arcs directly joining b to c (or d to e) In graph 2 there are two arcs directly joining v to w	B1 (A01.1) B1 (A01.1) [2]	Reason why graph 1 is not simple Reason why graph 2 is not simple	
		d	e.g. in graph 1 the vertex of degree 2 (a) is adjacent to each vertex of degree 4 (b and e), whereas in graph 2 the vertex of degree 2 (y) is adjacent to one vertex of degree 4 (z) but not the other (w) e.g. graph 1 has two pairs of vertices directly joined by two arcs (bc and de) whereas graph 2 only has one such pair (vw)	M1 (A02.3) A1(A01.1) [2]	Partially correct explanation of why the graphs are not isomorphic Fully correct explanation	
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Question			Answer/Indicative content	Marks	Guidance	
43		a	Let X play strategy P with probability p and strategy Q with probability $1 - p$ If Y plays strategy R, X can expect to win $4p - 3(1 - p) = 7p - 3$ If Y plays S, X can expect $-2p + (1 - p) = 1 - 3p$ $p = 0 \Rightarrow \min E(\text{win}) = -3$ $p = 1 \Rightarrow \min E(\text{win}) = -2$ $7p - 3 = 1 - 3p \Rightarrow p = 0.4$ and $\min E(\text{win}) = -0.2$ X should choose randomly between strategies P and Q so that P has prob 0.4 and Q has prob 0.6	M1 (A03.3) A1 (A01.1) E1 (A03.4) M1 (A01.1a) A1(A03.4) [5]	Calculating expected winnings if Y plays R or S Both correct Explicitly considering extreme points, or using a sketch graph to show that optimum point is at intersection Solving their expressions simultaneously to achieve their p Interpretation of $p = 0.4$ in context	May use p for P (X plays Q) May add a constant throughout Need not simplify expressions BC
		b	E.g. You must assume that Y is not going to play predictably E.g. You must assume that Y is not going to use a pure strategy	E1 (A03.5b) [1]	For one limitation of the model with reference to Y's limited behaviour	
Copyright			Total	6		



Question			Answer/Indicative content	Marks	Guidance	
44		a	3 possible orders: ABCD ACBD CABD	B1 (A01.1) [1]	Stating 3 or listing the three orders	
		b	E must come after C, but otherwise it can go anywhere If A, B, C, D are in the order A B C D then E can go immediately after C or D $\Rightarrow$ 2 If they are in the order A C B D then E can go immediately after C, B or D $\Rightarrow$ 3 If they are in the order C A B D then E can go immediately after C, A, B or D $\Rightarrow$ 4 In total there are 9 possible orders	B1 (A01.1) M1(A02.1) E1(A02.4) [3]	Seen, or implied by being true for all given orders Breaking the problem up into cases Showing that there are 9 orders and no more	ABCED ABCDE ACEBD ACBED ACBDE CEABD CAEBD CABED CABDE
		c	F must come after C If C is in the third position (A B C E D, A B C D E) there are 3 possibilities for F $\Rightarrow$ 6 If C is in the second position (A C E B D, A C B E D, A C B D E) there are 4 possibilities for F $\Rightarrow$ 12 If C is in the first position (C E A B D, C A E B D, C A B E D, C A B D E) there are 5 possibilities for F $\Rightarrow$ 20 In total there are 38 possible orders	B1 (A01.1) M1 (A02.1) E1 (A02.4) [3]	Seen, or implied by being true for all given orders Making a substantial start at counting the possibilities, this may involve starting again With appropriate working	A listing of the possibilities, without written reasoning, would score M0
			Total	7		



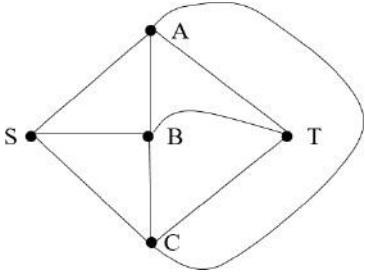
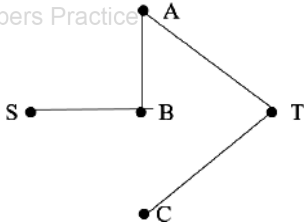
Question			Answer/Indicative content	Marks	Guidance																					
45			(A) e.g. Is this journey possible? e.g. Is it possible to travel from Edinburgh to Southampton, leaving Edinburgh after 9 am and arriving in Southampton by 4 pm?	B1 (AO1.1)	Any relevant problem that demonstrates the idea of existence	In the context of this train																				
			(B) e.g. What is the latest time I can leave Edinburgh to arrive in Southampton by 4 pm?	B1 (AO1.1)	Any relevant problem that demonstrates the idea of optimisation	In the context of this train journey																				
				[2]																						
			Total	2																						
46		a	<table><tr><td><i>N</i></td><td><i>K</i></td><td><i>M</i></td><td>Display</td></tr><tr><td>97 531</td><td>3</td><td></td><td></td></tr><tr><td>531</td><td>2</td><td>5</td><td>5</td></tr><tr><td>31</td><td>1</td><td>3</td><td>3</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr></table>	<i>N</i>	<i>K</i>	<i>M</i>	Display	97 531	3			531	2	5	5	31	1	3	3	1	0	1	1	M1(AO 1.1)	<i>N</i> = 531 and <i>K</i> = 2	
			<i>N</i>	<i>K</i>	<i>M</i>	Display																				
			97 531	3																						
			531	2	5	5																				
			31	1	3	3																				
1	0	1	1																							
M1(AO 1.1)	<i>N</i> = 31 and <i>K</i> = 1																									
A1(AO 1.1)	<i>M</i> (or display) = 5, 3, 1																									
				[3]																						
		b	<i>K</i> is initially 3 and is reduced by 1 in each pass, so it will reach 0 and the algorithm will stop.	E1(AO 2.4)	Evidence of understanding that <i>K</i> acts as a counter.																					
				[1]																						
Total				4																						



Question			Answer/Indicative content	Marks	Guidance	
47			x = area (m^2) decorated using paint y = area (m^2) decorated using panelling z = area (m^2) decorated using wallpaper Minimise ($C =$) $1.12x + 4.62y + 1.61z$ Subject to $0.50x + 0.34y + 0.28z \leq 8$ $x + y + z \geq 30$ and $x \geq 0, y \geq 0, z \geq 0$	B1(AO 3.1) M1(AO 3.3) M1(AO 3.3) A1(AO 1.1) B1(AO 1.1) B1(AO 1.1)	Defining appropriate variables for the areas, with units Min (their) C (their) $T \leq 8$ Both correct Sum ≥ 30 Non-negativity	
			Total	6		



Question			Answer/Indicative content	Marks	Guidance																			
48		a	Activity A(5) has float 0. Activity B(6) has float 0. Activity C(8) has float 13. Activity D(5) has float 11. Activity E(4) has float 6. Activity F(2) has float 15. Activity G(4) has float 13. Activity H(3) has float 17.	M1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [3]	Forward pass with at most 1 independent error Backward pass with at most 1 independent error Both passes correct	From minimum project duration from forward pass																		
		b	<table border="1"><thead><tr><th>Activity</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th></tr></thead><tbody><tr><td>Float</td><td>0</td><td>3</td><td>0</td><td>4</td><td>3</td><td>4</td><td>0</td><td>4</td></tr></tbody></table>	Activity	A	B	C	D	E	F	G	H	Float	0	3	0	4	3	4	0	4	B1(AO 1.2) M1(AO 1.1) A1(AO 1.1) [3]	Critical activities A, C, G = 0 At least two of B, E, F (ft (i)) Correct for all non-critical (cao)	
Activity	A	B	C	D	E	F	G	H																
Float	0	3	0	4	3	4	0	4																
		c	$4 < x < 7$	M1(AO 2.2a) A1ft(AO 2.2a) [2]	4 + ft(ii) seen cao	Or ft(i) (their) 13 – (their) 6																		
			Total	8																				

Question			Answer/Indicative content	Marks	Guidance																																					
49		a		B1(AO 3.3) [1]	This graph (K_5 but missing arc ST)	Graph may be drawn with arcs crossing																																				
		b	(i) Shortest path may not pass through A, B and C.	B1(AO 3.5b) [1]	May miss out some of A, B, C	May miss out some vertices																																				
		b	(ii) Minimum spanning tree may have branches and not be a path (from S to T)	B1(AO 3.5b) [1]	Tree may not be a path	Tree may be a path but not have S and T as its ends																																				
		c	e.g. <table><tr><th></th><th>1 S</th><th>3 A</th><th>2 B</th><th>5 C</th><th>4 T</th></tr><tr><th>S</th><td>0</td><td>3</td><td>2</td><td>5</td><td>4</td></tr><tr><th>A</th><td>3</td><td>0</td><td>2</td><td>2.5</td><td>2</td></tr><tr><th>B</th><td>2</td><td>2</td><td>0</td><td>3.2</td><td>2.5</td></tr><tr><th>C</th><td>5</td><td>2.5</td><td>3.2</td><td>0</td><td>2</td></tr><tr><th>T</th><td>4</td><td>2</td><td>2.5</td><td>2</td><td>0</td></tr></table>  Total weight = 8 (km) SB = 2 BA = 2 AT = 2 TC = 2		1 S	3 A	2 B	5 C	4 T	S	0	3	2	5	4	A	3	0	2	2.5	2	B	2	2	0	3.2	2.5	C	5	2.5	3.2	0	2	T	4	2	2.5	2	0	M1(AO 3.1b) A1(AO 1.1) A1(AO 1.1) [3]	Prim may start from any vertex Prim's algorithm (either version) or Kruskal's algorithm, may be implied from a correct tree Correct tree drawn or listed 8 seen as total weight	Kruskal: SB = 2 AB = 2 AT = 2 CT = 2 in any order
	1 S	3 A	2 B	5 C	4 T																																					
S	0	3	2	5	4																																					
A	3	0	2	2.5	2																																					
B	2	2	0	3.2	2.5																																					
C	5	2.5	3.2	0	2																																					
T	4	2	2.5	2	0																																					



Question			Answer/Indicative content	Marks	Guidance
		d	S - B - A - C - T = 8.5 km	B1(AO 3.4) B1(AO 3.2a) [2]	Any route from S to T via A, B, C 8.5 km, cao (including units)
			Total	8	



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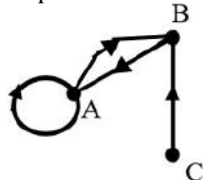


Question			Answer/Indicative content	Marks	Guidance																										
50		a	A B <table><tr><td></td><td>red 1</td><td>red 2</td><td>blue 3</td><td>blue 4</td></tr><tr><td>red 1</td><td>0</td><td>2</td><td>-4</td><td>-5</td></tr><tr><td>blue 2</td><td>3</td><td>0</td><td>6</td><td>8</td></tr><tr><td>red 3</td><td>-3</td><td>-6</td><td>0</td><td>-7</td></tr><tr><td>blue 4</td><td>5</td><td>6</td><td>-12</td><td>0</td></tr></table>		red 1	red 2	blue 3	blue 4	red 1	0	2	-4	-5	blue 2	3	0	6	8	red 3	-3	-6	0	-7	blue 4	5	6	-12	0	M1(AO 1.1a) M1(AO 1.1) A1(AO 1.1) [3]	At least two non-zero entries for same colours correct At least two non-zero entries for different colours correct All correct	
	red 1	red 2	blue 3	blue 4																											
red 1	0	2	-4	-5																											
blue 2	3	0	6	8																											
red 3	-3	-6	0	-7																											
blue 4	5	6	-12	0																											
		b	row minima -5, 0, -7, -12 maximin = 0 col maxima 5, 6, 6, 8 minimax = 5 Play-safe for A is blue 2 Play-safe for B is red 1	M1(AO 1.1a) A1ft(AO 1.1) [2]	Row minima and col maxima seen in text or margins of table Both play-safe strategies from correct working seen (ft their matrix)	May use negatives of col maxima																									
		c	Dominated by blue 2 3 > -3, 0 > -6, 6 > 0, 8 > -7	E1(AO 2.1) [1]	In each column entry in row blue 2 ≥ entry in row red 3 and they are not all equal	Row blue 2 gives a higher pay-off than row red 3 in each column																									
		d	0 ≠ 5 so unstable	E1(AO 2.4) [1]	Or 0 + (-5) ≠ 0 so unstable Or max(row min) ≠ min(col max) Need to see reason as well as correct conclusion (ft their matrix)	Or in words, e.g. unstable since if both play-safe A would do better by changing to blue 4 (or B would do better by changing to red 2)																									
			Total	7																											



Question			Answer/Indicative content	Marks	Guidance	
51		a	After 1 st pass 15 12 23 9 14 16 27 31 4 After 2 nd pass 15 23 12 14 16 27 31 9 4 After 3 rd pass 23 15 14 16 27 31 12 9 4	B1(AO 1.1) B1(AO 1.1) [2]	FT	Allow one error in 1 st pass or 2 nd pass
		b	After 1 st pass 23 15 16 27 31 14 12 9 4 After 2 nd pass 23 16 15 27 31 14 12 9 4 After 3 rd pass 27 23 16 15 31 14 12 9 4	M1(AO 1.1) A1(AO 1.1) [2]	15 and 16 swapped Correct list after 3 rd pass	Missing out 1 st pass and giving sorted list as after 3 rd pass = M1 A0
		c	The last 4 numbers were already fixed from bubble sort, so Sophie only needs to sort the first 5 masses. For a list of length 5, shuttle sort uses 4 passes, so only 1 more pass is needed	E1(AO 2.4) E1(AO 2.2a) [2]	4 numbers fixed by bubble sort or 4 passes of bubble sort 5 – 1 = 4 passes of shuttle needed, three have already been done	She has already done 7 passes (4 of bubble and 3 of shuttle) = E1 E1 “List is sorted after 1 more pass” = E0 E0
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Question			Answer/Indicative content	Marks	Guidance	
52		a	(i) In-degrees $A = 2, B = 2, C = 0$ (column sums) Out-degrees $A = 2, B = 1, C = 1$ (row sums)	M1(AO 1.1) A1(AO 1.1) [2]	Any three correct All six correct	
		a	(ii) start C: outdegree = 1 + indegree end B: indegree = outdegree + 1 others A: indegree = outdegree	M1(AO 1.1) A1(AO 1.2) [2]	Dealing with any one of start/end/others in terms of relationship between indegree and outdegree All three correct in words	Alternative SC1 Graph seen or implied  Semi-Eulerian trail $C - B - A - A - B$
		b	(i) Matrix is symmetric about the diagonal DD, EE, FF	E1(AO 2.5) [1]	Describing symmetry about lead diagonal (in words or numerically)	
		b	(ii) Vertex degrees are given by row sums (or column sums) $D = 3, E = 3, F = 2$ Exactly two odd vertices, so semi-Eulerian	E1(AO 2.1) B1(AO 1.2) [2]	Degree = row sum (or col sum) No reference to using row sums (or col sums) = E0 Two odd vertices and 1 even D and E odd (or 2 odd) without evidence of 'exactly' = E0	D and E have odd row sums and F has even row sum



Question			Answer/Indicative content	Marks	Guidance	
		c	In digraph loop contributes 1 to in-degree and 1 to out-degree of A In undirected graph loop contributes 2 to degree of D	E1(AO 2.1) B1(AO 1.1) [2]	1 in and 1 out, digraph or directed 2 arc endings, undirected	Degree = in-degree + out-degree
			Total	9		



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Question			Answer/Indicative content	Marks	Guidance	
53		a	$\{A B C D\}$ $\{A B C\} \{D\}$ $\{A B D\} \{C\}$ $\{A C D\} \{B\}$ $\{B C D\} \{A\}$ $\{A B\} \{C D\}$ $\{A C\} \{B D\}$ $\{A D\} \{B C\}$ $\{A B\} \{C\} \{D\}$ $\{A C\} \{B\} \{D\}$ $\{A D\} \{B\} \{C\}$ $\{B C\} \{A\} \{D\}$ $\{B D\} \{A\} \{C\}$ $\{C D\} \{A\} \{B\}$ $\{A\} \{B\} \{C\} \{D\}$	B1(AO 3.1b) B1(1.1) B1(1.1)	Four different partitions of the form $\{XXX\} \{X\}$ or $XXX X$ Three different partitions of the form $\{XX\} \{XX\}$ or $XX XX$ Six different partitions of the form $\{XX\} \{X\} \{X\}$ or $XX X X$	Order of elements within sets does not matter but sets in a partition must be obviously joined together Sets must be non-empty May use any unambiguous notation
		b	7	B1ft(1.1) [1]	Correct or follow through from (a)	



Question			Answer/Indicative content	Marks	Guidance	
		c	A can go in either taxi but then B must be in the other, D can go in either taxi which gives 2×24	E1(AO 2.4) B1(AO 1.1) [2]	Using multiplication principle or listing possibilities 4 cao	Explanation or 2×2 must be seen not implied from answer 4

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Question			Answer/Indicative content	Marks	Guidance	
		d	Two taxis: 4, 2 or 3, 3: ${}^6C_4 + {}^6C_3/2! = 15+10 = 25$ Three taxis: 4, 1, 1 or 3, 2, 1 or 2, 2, 2 $= {}^6C_4 + {}^6C_3 \times {}^3C_2 + ({}^6C_2 \times {}^4C_2 \div 3!)$ $= 15 + 60 + 15 = 90$ Four taxis: 3, 1, 1, 1 or 2, 2, 1, 1 $= {}^6C_3 + ({}^6C_2 \times {}^4C_2 \div 2!) = 20 + 45 = 65$ Five taxis: 2, 1, 1, 1, 1 = ${}^6C_2 = 15$ Six taxis: 1, 1, 1, 1, 1, 1 = 1 $1 + 15 + 65 + 90 + 25 = 196$	M1(AO 1.1) M1(AO 3.1b) M1(AO 3.1b) M1(AO 1.1) M1dep (AO 2.1) A1(AO 2.2a) [6]	Using 6C_r with $r = 2, 3$ or 4 Product ${}^6C_r \times {}^{6-r}C_{5-r}$ with $r = 2$ or 3 Dealing with 2, 2, 2 case Dealing with 2, 2, 1, 1 case 1+15+(their)65+(their)90+(their)25 dependent on all previous M marks 196 cao	Or 20×3 or 15×4 or 60 Alternative: Identifying that there are 203 partitions M1 A1 exclude one taxi =1B1 two taxis: 5, 1=6B1 (their) 203 - 1 - 6 = 196 M1 A1
			Total	12		